

317521: Computer Networks

Unit Number: 4

Unit Name: Transport Layer

Unit Outcomes: CO4

CO4: Implement client-server applications using sockets

Syllabus

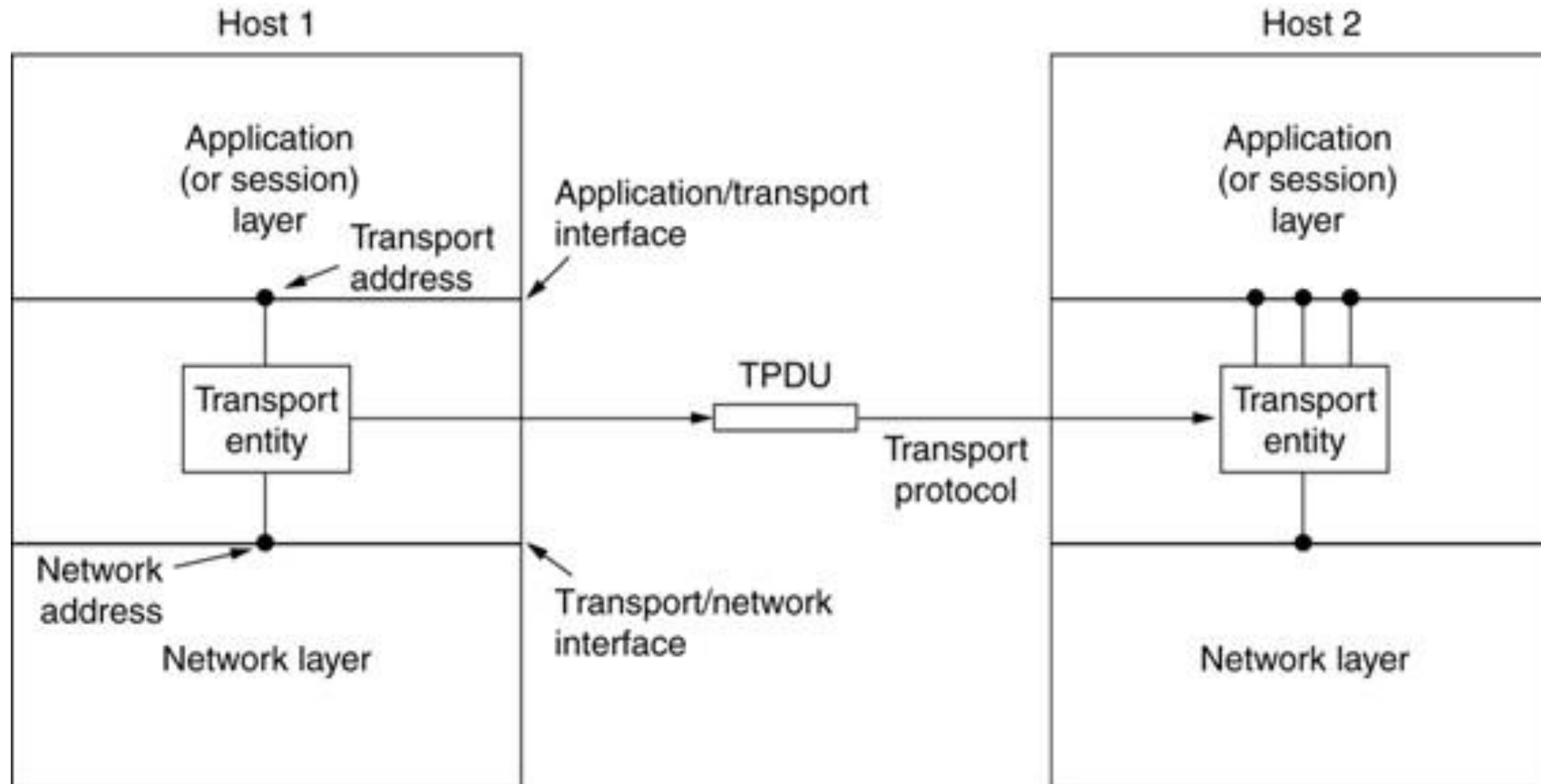
Process to Process Delivery, Services, Socket Programming.

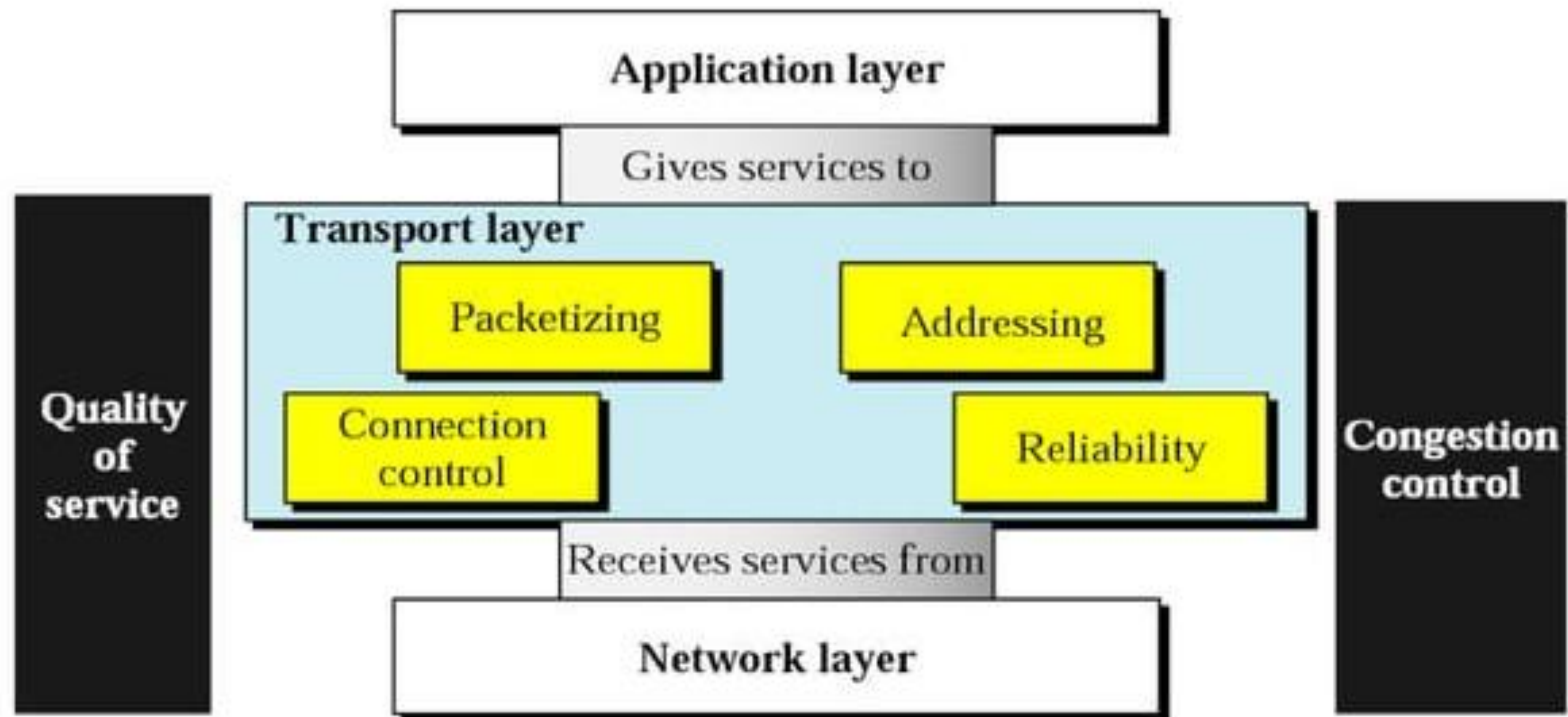
Elements of Transport Layer Protocols: Addressing, Connection establishment, Connection release, Flow control and buffering, Multiplexing, Congestion Control.

Transport Layer Protocols: TCP and UDP, SCTP, RTP, Congestion control and Quality of Service (QoS), Differentiated services, TCP and UDP for Wireless networks.

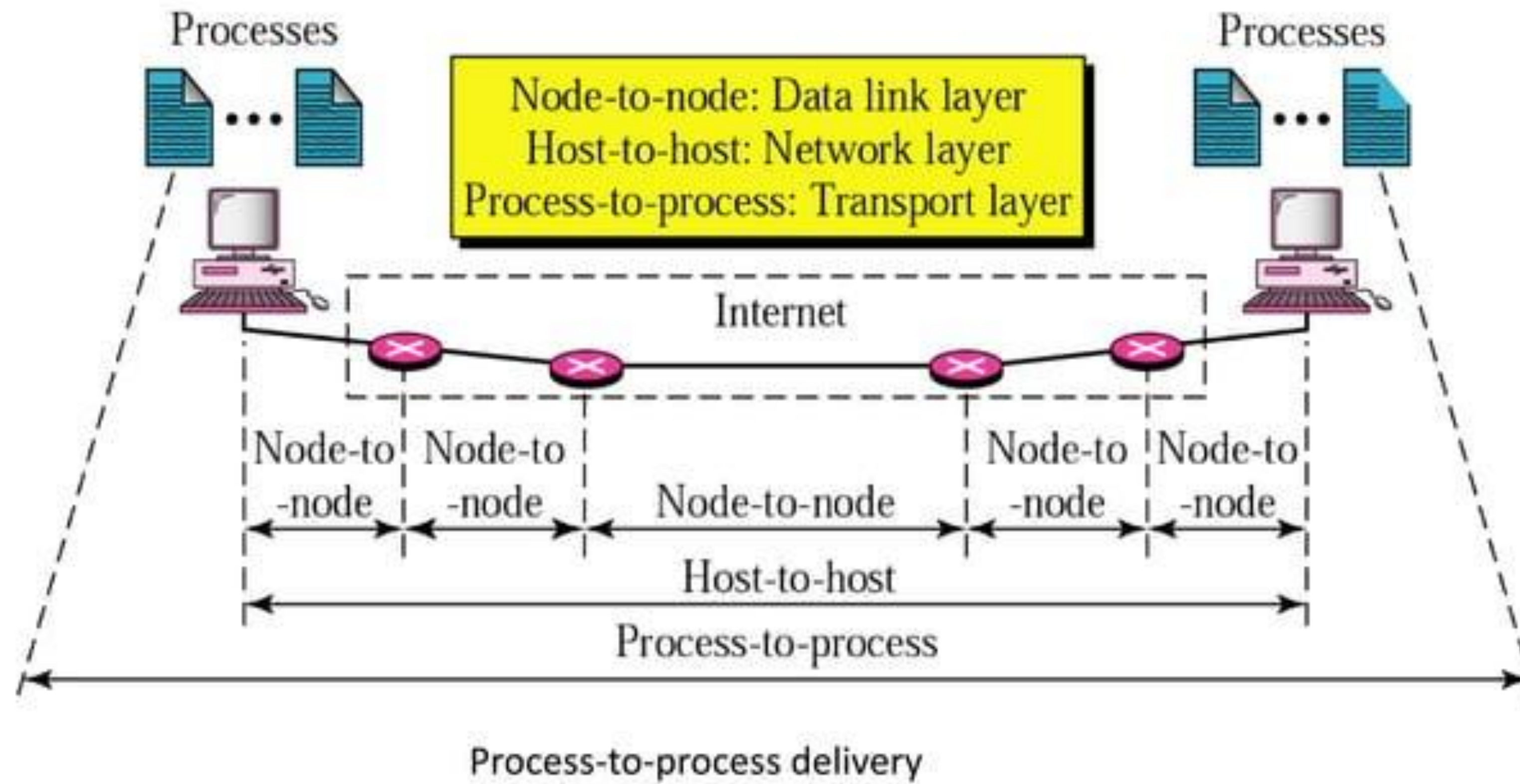
Transport layer

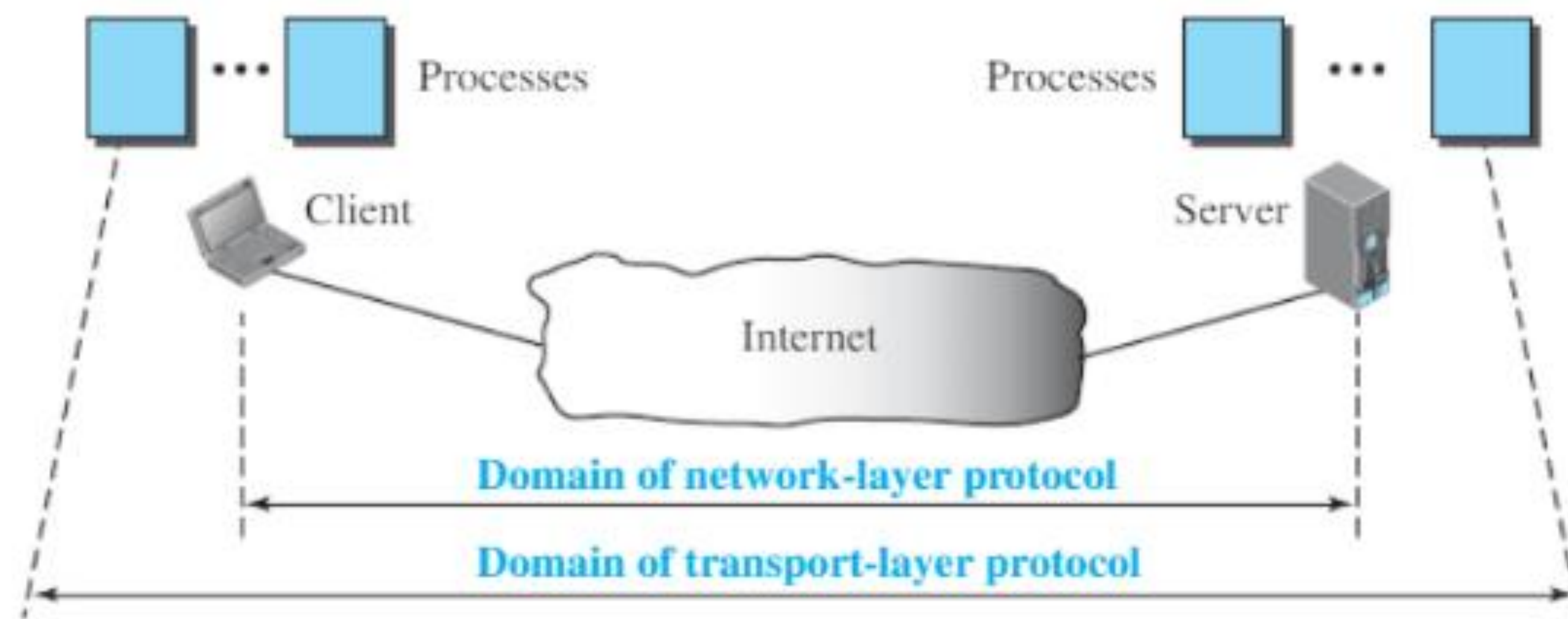
- The network layer exists on end hosts and routers in the network.
- The end-user cannot control what is in the network.
- So the end-user establishes another layer, only at end hosts, to provide a transport service
- While the network layer deals with only a few transport entities, the transport layer allows several concurrent applications to use the transport service.
- It provides a common interface to application writers.
- In essence, an application writer can write code once using the transport layer primitive and use it on different networks (but with the same transport layer).





Transport Layer





Transport Service Primitives

Primitive	Packet sent	Meaning
LISTEN	(none)	Block until some process tries to connect
CONNECT	CONNECTION REQ.	Actively attempt to establish a connection
SEND	DATA	Send information
RECEIVE	(none)	Block until a DATA packet arrives
DISCONNECT	DISCONNECTION REQ.	This side wants to release the connection

Elements of Transport Protocols

- Addressing,
- Connection Establishment,
- Connection Release,
- Flow Control and Buffering,
- Multiplexing,
- Crash Recovery

Addressing

- To achieve process-to-process communication, the most common is through the client-server paradigm.
- For communication, we must define the local host, local process, remote host, and remote process.
- The local host and the remote host are defined using IP addresses.
- To define the processes, we need second identifiers, called port numbers.
- In the TCP/IP protocol suite, the port numbers are integers between 0 and 65,535 (16 bits)

ICANN has divided the port numbers into three ranges: well-known, registered, and dynamic (or private)

Well-known ports- The ports ranging from 0 to 1023 are assigned and controlled by ICANN. These are the well-known ports.

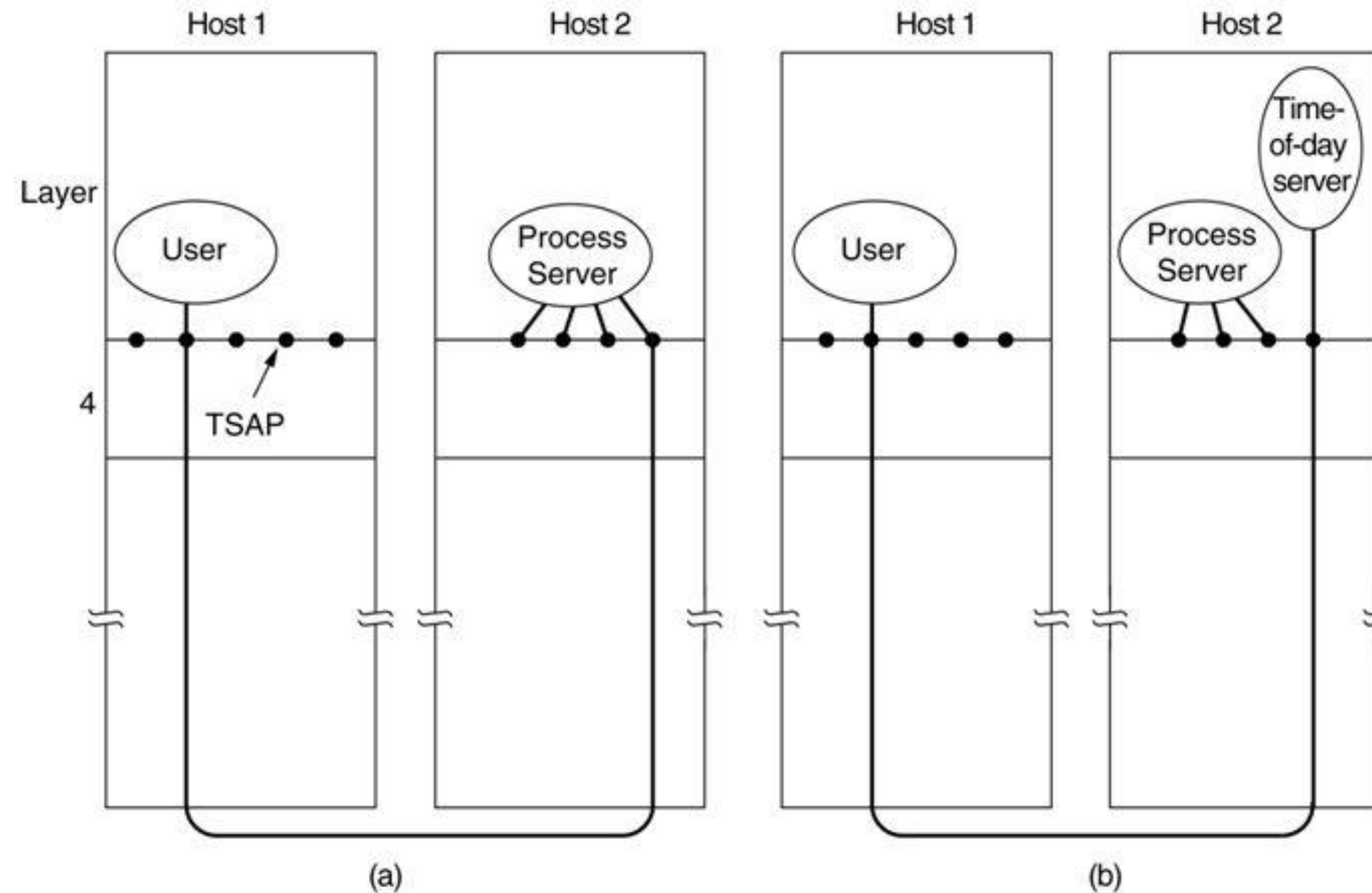
Registered ports-The ports ranging from 1024 to 49,151 are not assigned or controlled by ICANN. They can only be registered with ICANN to prevent duplication.

Dynamic ports-The ports ranging from 49,152 to 65,535 are neither controlled nor registered. They can be used as temporary or private port numbers.

Connection Establishment

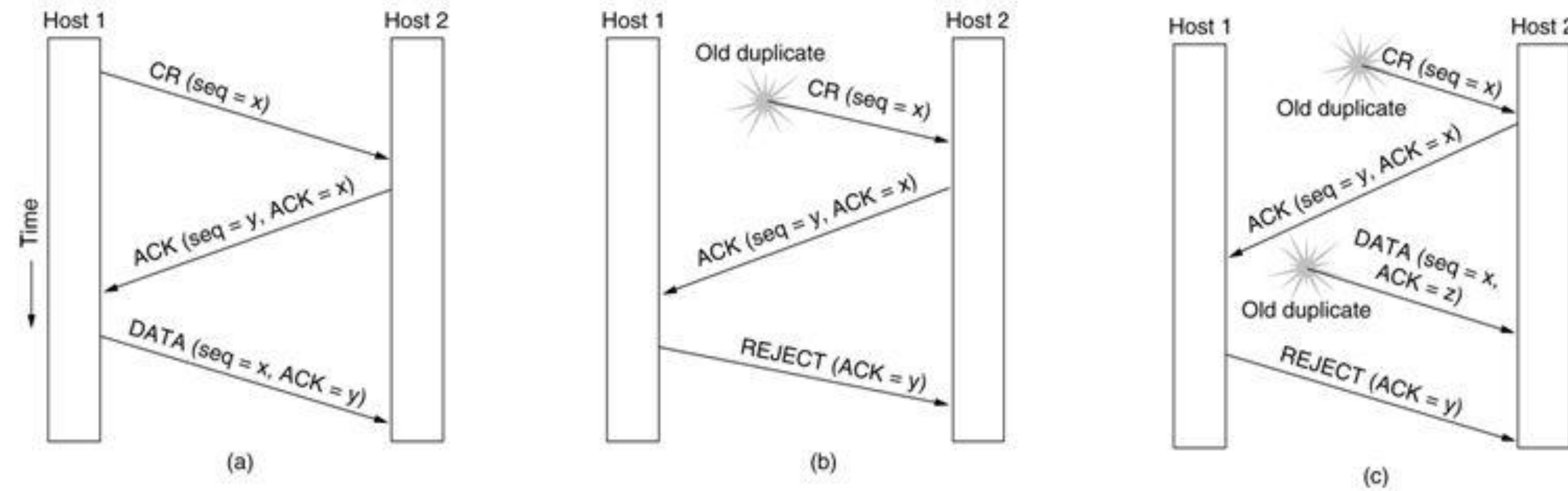
How a user process in host 1 establishes a connection with a time-of-day server in host 2.

Connection Establishment (1)



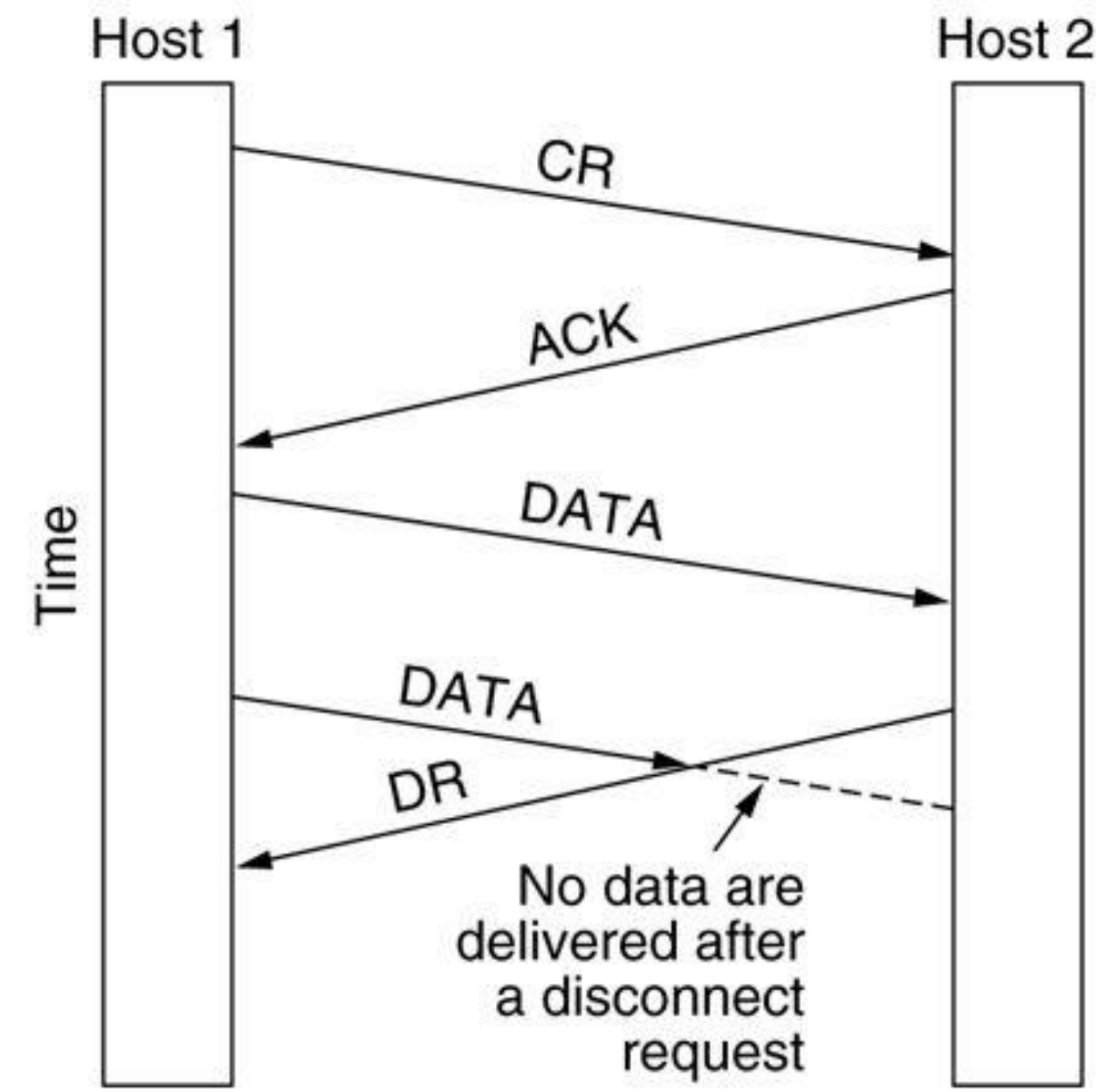
How a user process in host 1 establishes a connection with a time-of-day server in host 2.

Connection Establishment (2)



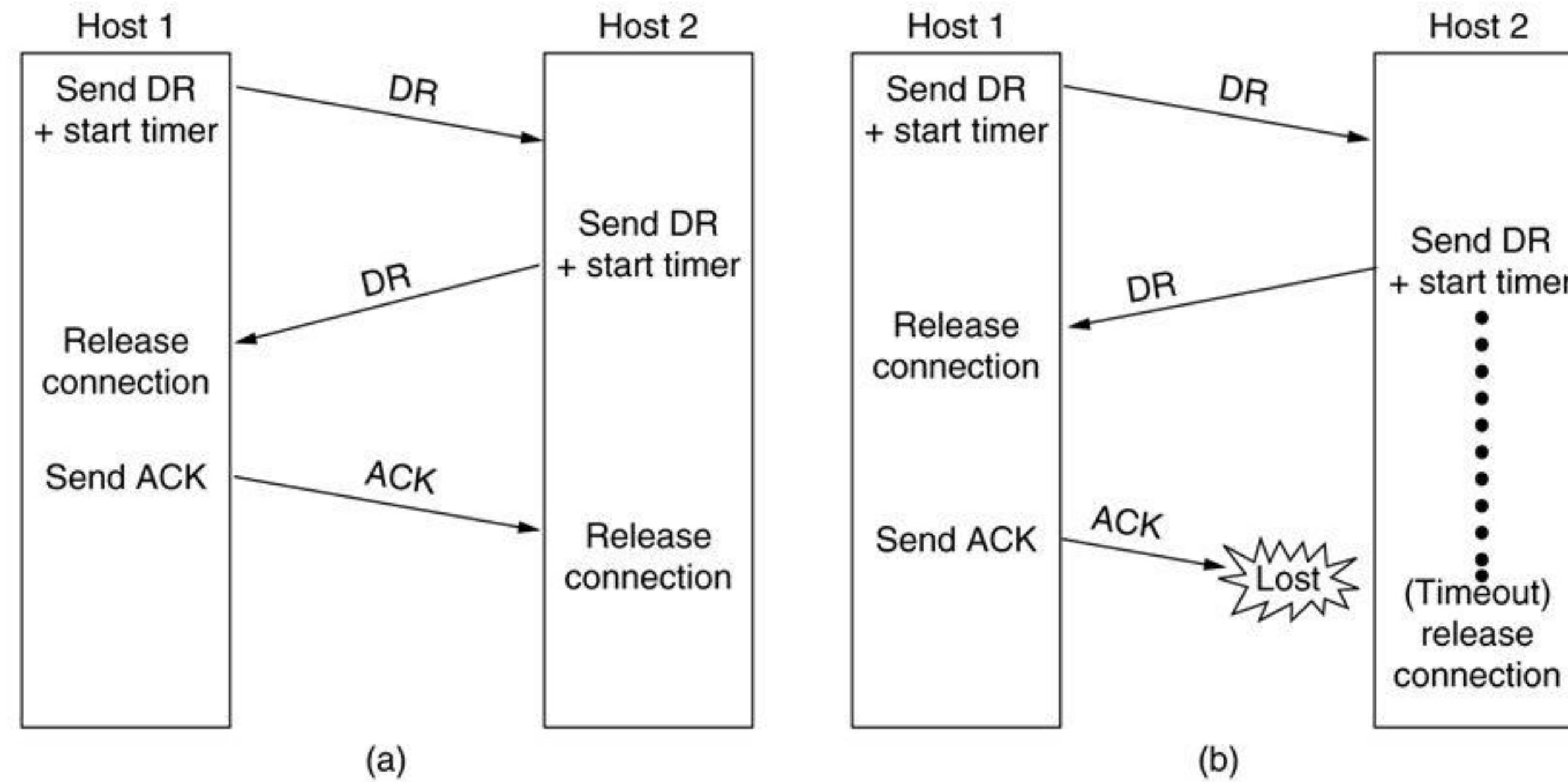
- Three protocol scenarios for establishing a connection using a three-way handshake. CR denotes CONNECTION REQUEST.
- (a) Normal operation,
 - (b) Old CONNECTION REQUEST appearing out of nowhere.
 - (c) Duplicate CONNECTION REQUEST and duplicate ACK.

Connection Release



Abrupt disconnection with loss of data.

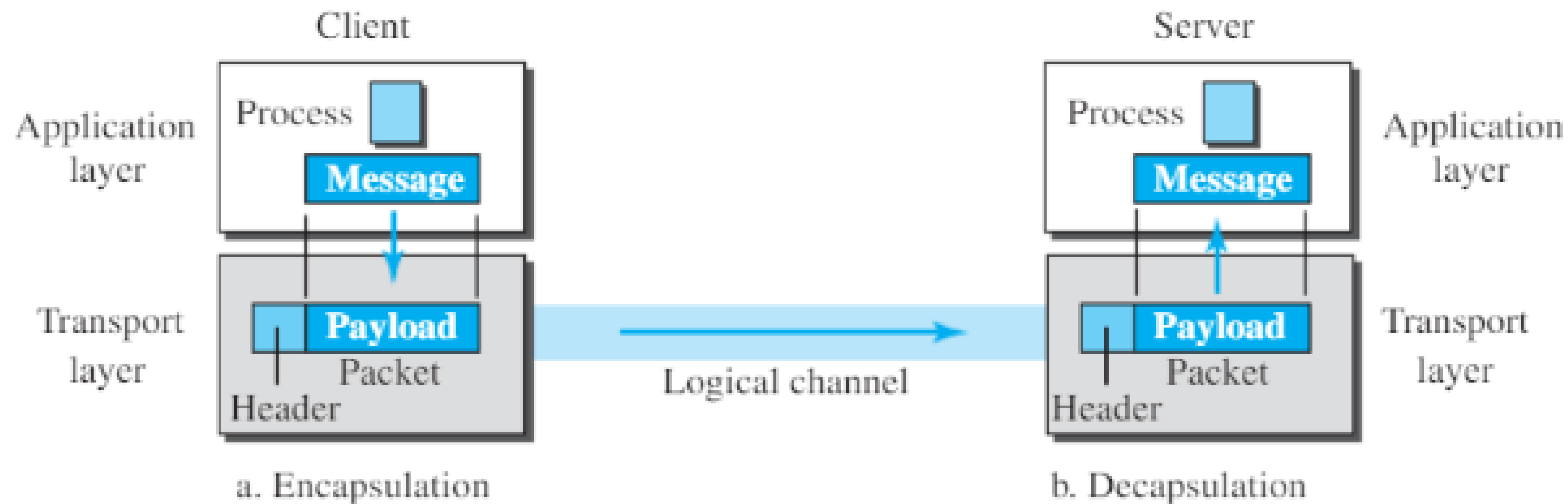
Connection Release (3)



Four protocol scenarios for releasing a connection. (a) Normal case of a three-way handshake. (b) final ACK lost.

Encapsulation and Decapsulation

To send a message from one process to another, the transport-layer protocol encapsulates and decapsulates messages.



Multiplexing and Demultiplexing

Flow Control

The transport layer provides a flow control mechanism between the adjacent layers of the TCP/IP model. TCP also prevents data loss due to a fast sender and slow receiver by imposing some flow control techniques. It uses the method of sliding window protocol which is accomplished by the receiver by sending a window back to the sender informing the size of data it can receive.

Figure 23.9 *Pushing or pulling*

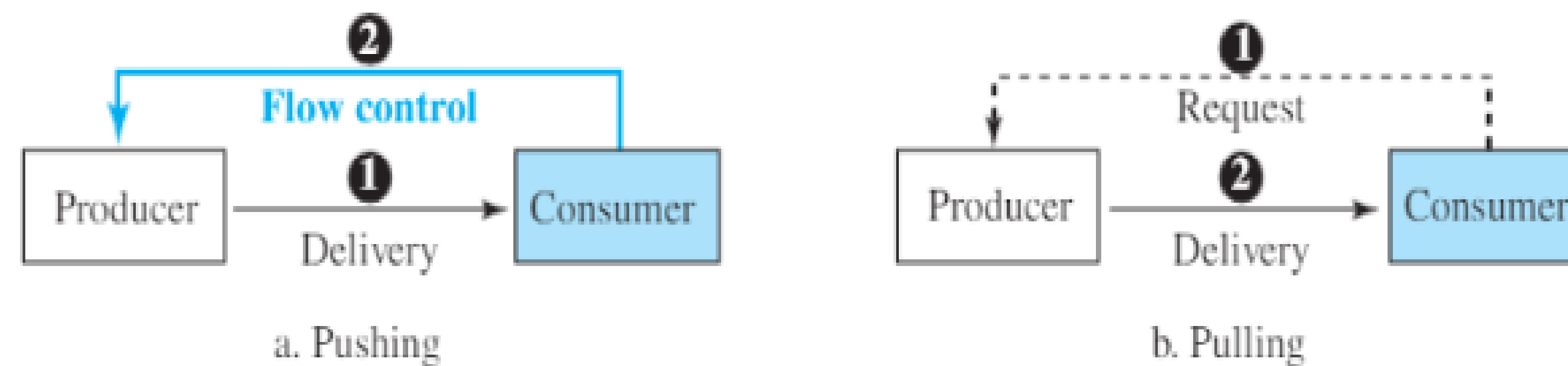
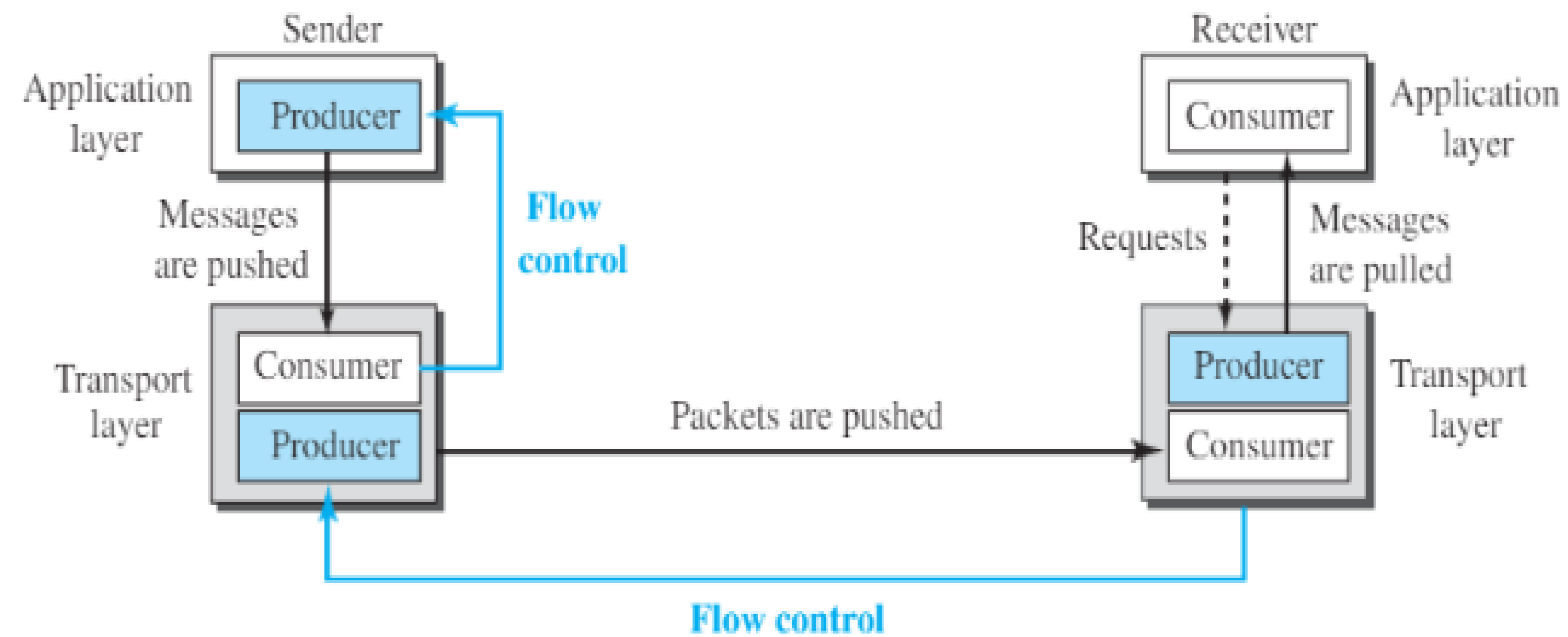


Figure 23.10 *Flow control at the transport layer*



Error control

Transport-Layer Services Error control In the Internet, since the underlying network layer (IP) is unreliable, we need to make the transport layer reliable if the application requires reliability.

Reliability can be achieved to add error control services to the transport layer.

Error control at the transport layer is responsible for

1. Detecting and discarding corrupted packets.
2. Keeping track of lost and discarded packets and resending them.
3. Recognizing duplicate packets and discarding them.
4. Buffering out-of-order packets until the missing packets arrive.

Transport-Layer Services Sequence numbers, Error control requires that the sending transport layer knows which packet is to be resent and the receiving transport layer knows which packet is a duplicate, or which packet has arrived out of order. This can be done if the packets are numbered.

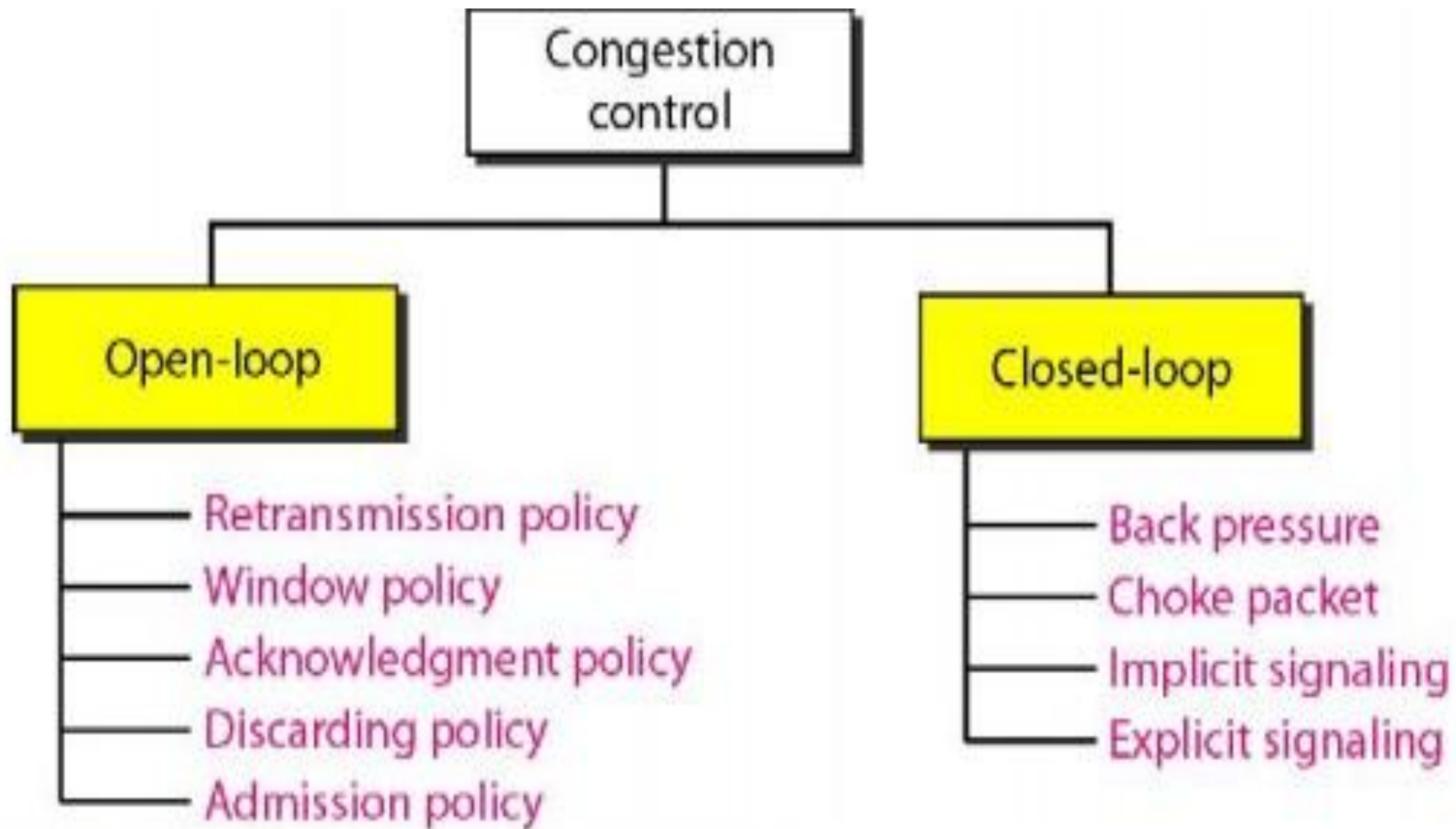
We can add a field to the transport-layer packet to hold the sequence number of the packet

Congestion Control

- Congestion occurs when the network traffic is too heavy. i.e. when the number of packets sent to the network is much greater than the capacity of the network.
- The mechanisms used to control or handle congestion in a network are called congestion control mechanisms.
- The main objective of congestion control is to reduce congestion in the network, and ensure the delivery of packets to the respective destinations.

There are two ways of congestion control:

- Open loop congestion control (congestion prevention)
- Closed loop congestion control (congestion removal)



1. Open-Loop Congestion Control

In open-loop congestion control, policies are applied to prevent congestion before it happens. In these mechanisms, congestion control is handled by either the source or the destination.

a. Retransmission Policy

- Retransmission is sometimes unavoidable. If the sender feels that a sent packet is lost or corrupted, the packet needs to be retransmitted.
- Retransmission in general may increase congestion in the network. However, a good retransmission policy can prevent congestion.
- The retransmission policy and the retransmission timers must be designed to optimize efficiency and at the same time prevent congestion.
- For example, the retransmission policy used by TCP is designed to prevent or alleviate congestion.

b. Window Policy

- The type of window at the sender may also affect congestion. The Selective Repeat window is better than the Go-Back-N window for congestion control.
- In the Go-Back-N window, when the timer for a packet times out, several packets may be resent, although some may have arrived safe and sound at the receiver. This duplication may make the congestion worse.
- The Selective Repeat window, on the other hand, tries to send the specific packets that have been lost or corrupted.

c. Acknowledgment Policy

- The acknowledgment policy imposed by the receiver may also affect congestion. If the receiver does not acknowledge every packet it receives, it may slow down the sender and help prevent congestion. Several approaches are used in this case.
- A receiver may send an acknowledgment only if it has a packet to be sent or a special timer expires. A receiver may decide to acknowledge only N packets at a time.
- We need to know that the acknowledgments are also part of the load in a network. Sending fewer acknowledgments means imposing fewer loads on the network.

d. Discarding Policy

- A good discarding policy by the routers may prevent congestion and at the same time may not harm the integrity of the transmission.
- For example, in audio transmission, if the policy is to discard less sensitive packets when congestion is likely to happen, the quality of sound is still preserved and congestion is prevented or alleviated.

e. Admission Policy

- An admission policy, which is a quality-of-service mechanism, can also prevent congestion in virtual-circuit networks. Switches in a flow, first check the resource requirement of a flow before admitting it to the network.
- A router can deny establishing a virtual circuit connection if there is congestion in the network or if there is a possibility of future congestion.

2. Closed-Loop Congestion Control

Closed-loop congestion control mechanisms try to alleviate congestion after it happens. Several mechanisms have been used by different protocols.

a. Backpressure

The technique of backpressure refers to a congestion control mechanism in which a congested node stops receiving data from the immediate upstream node or nodes. This may cause the upstream node or nodes to become congested, and they, in turn, reject data from their upstream nodes or nodes. And so on. Backpressure is a node-to-node congestion control that starts with a node and propagates, in the opposite direction of data flow, to the source. The backpressure technique can be applied only to virtual circuit networks, in which each node knows the upstream node from which a flow of data is coming. Figure 4.28 shows the idea of backpressure.

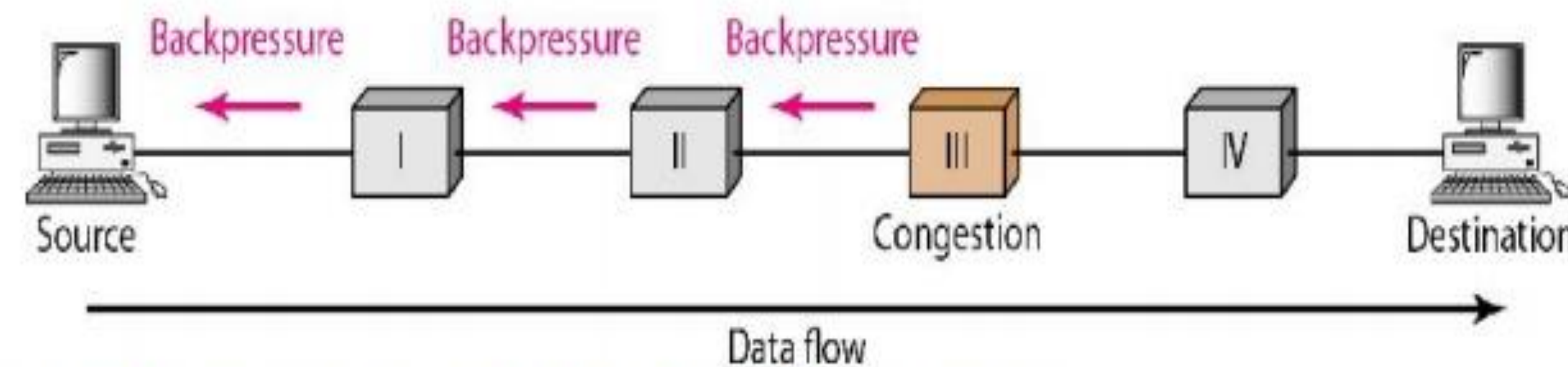


Figure 4.28 Backpressure method for alleviating congestion

Node III in the figure has more input data than it can handle. It drops some packets in its input buffer and informs node II to slow down. Node II, in turn, may be congested because it is slowing down the output flow of data. If node II is congested, it informs node I to slow down, which in turn may create congestion. If so, node I inform the source of data to slow down. This, in time, alleviates the congestion. Note that the pressure on node III is moved backward to the source to remove the congestion.

b. Choke Packet

- A choke packet is a packet sent by a node to the source to inform it of congestion. Note the difference between the backpressure and choke packet methods.
- In backpressure, the warning is from one node to its upstream node, although the warning may eventually reach the source station.
- In the choke packet method, the warning is from the router, which has encountered congestion, to the source station directly.
- The warning message goes directly to the source station; the intermediate routers, and does not take any action. Figure 4.29 shows the idea of a choke packet.

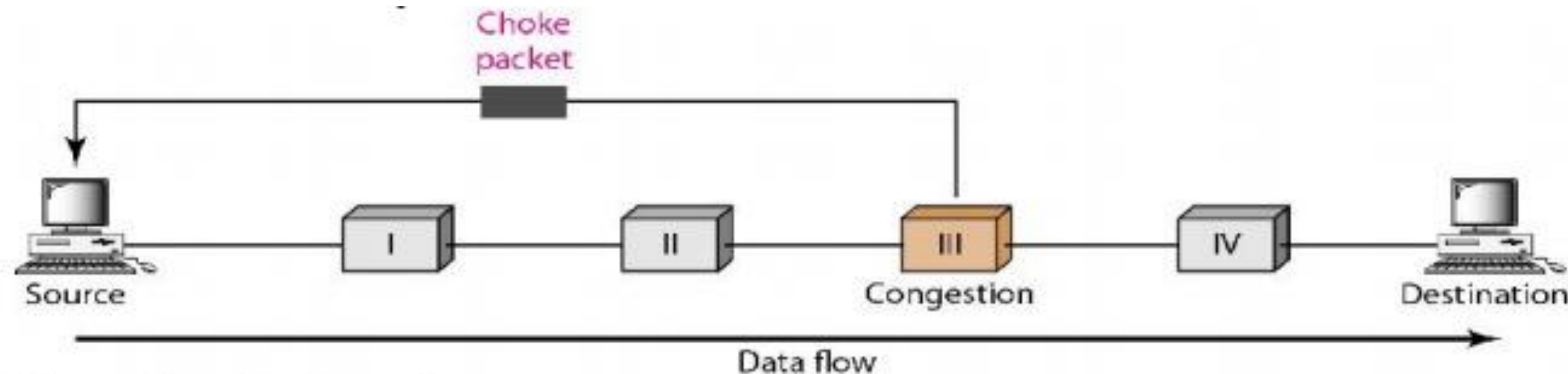


Figure 4.29 Choke packet

c. Implicit Signaling:

- In implicit signaling, there is no communication between the congested node or nodes and the source. The source guesses that there is congestion somewhere in the network from other symptoms.
- For example, when a source sends several packets and there is no acknowledgment for a while, one assumption is that the network is congested.
- The delay in receiving an acknowledgment is interpreted as congestion in the network; the source should slow down.

d. Explicit Signaling

- The node that experiences congestion can explicitly send a signal to the source or destination. The explicit signaling method, however, is different from the choke packet method.
- In the choke packet method, a separate packet is used for this purpose; in the explicit signaling method, the signal is included in the packets that carry data.
- Explicit signaling, can occur in either the forward or the backward direction.

i. Backward Signaling

A bit can be set in a packet moving in the direction opposite to the congestion. This bit can warn the source that there is congestion and that it needs to slow down to avoid the discarding of packets.

ii. Forward Signaling

A bit can be set in a packet moving in the direction of the congestion. This bit can warn the destination that there is congestion. The receiver in this case can use policies, such as slowing down the acknowledgments, to alleviate the congestion.

Effects of Congestion Control in Computer Network

- Improved Network Stability:** Congestion control helps keep the network stable by preventing it from getting overloaded. It manages the flow of data so the network doesn't crash or fail due to too much traffic.
- Reduced Latency and Packet Loss:** Without congestion control, **data transmission** can slow down, causing delays and data loss. Congestion control helps manage traffic better, reducing these delays and ensuring fewer data packets are lost, making data transfer faster and the network more responsive.
- Enhanced Throughput:** By avoiding congestion, the network can use its resources more effectively. This means more data can be sent in a shorter time, which is important for handling large amounts of data and supporting high-speed applications.

- Fairness in Resource Allocation:** Congestion control ensures that network resources are shared fairly among users. No single user or application can take up all the bandwidth, allowing everyone to have a fair share.
- Better User Experience:** When data flows smoothly and quickly, users have a better experience. Websites, online services, and applications work more reliably and without annoying delays.
- Mitigation of Network Congestion Collapse:** Without congestion control, a sudden spike in data traffic can overwhelm the network, causing severe congestion and making it almost unusable. Congestion control helps prevent this by managing traffic efficiently and avoiding such critical breakdowns.

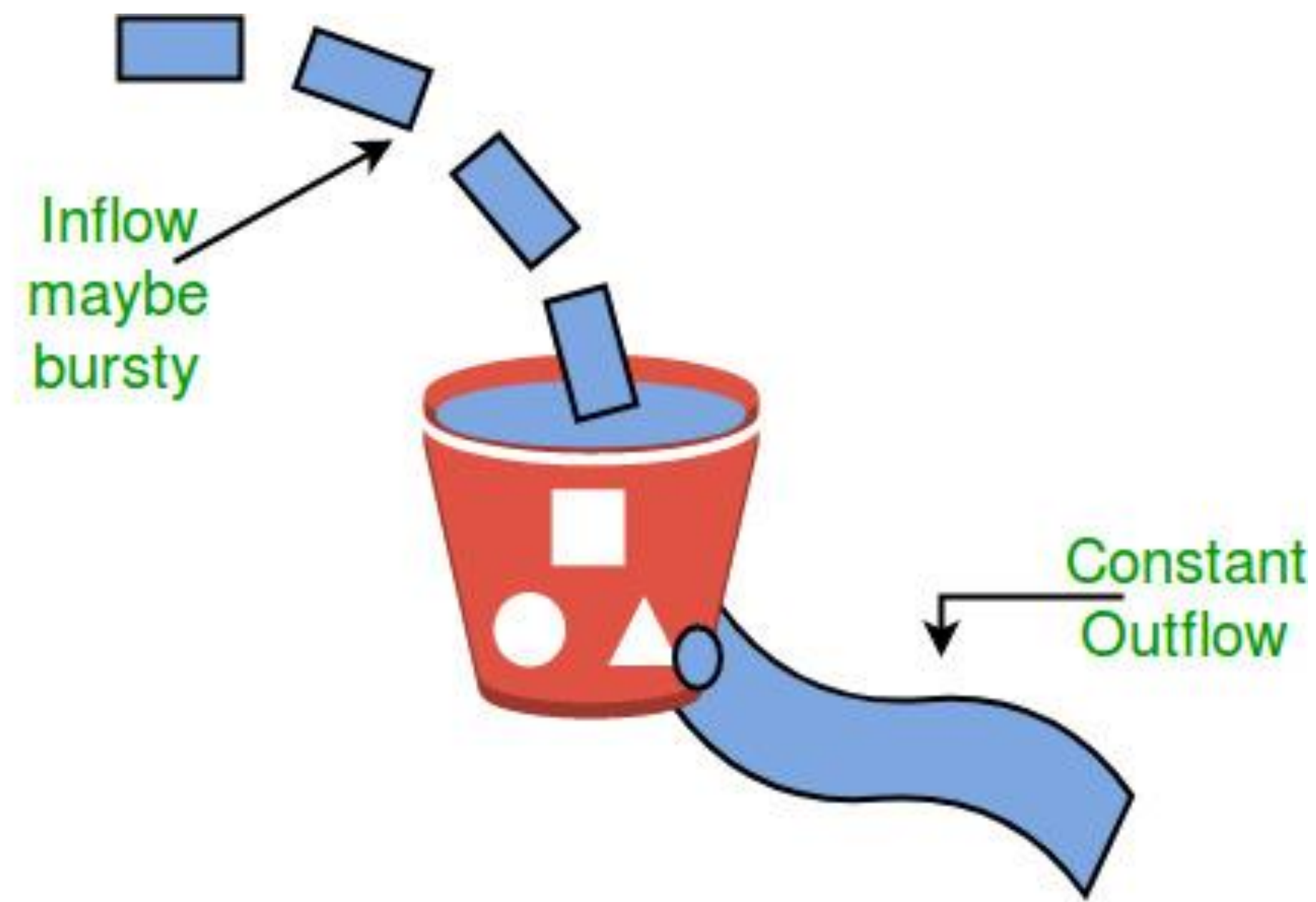
Congestion Control Algorithm

- Congestion Control is a mechanism that controls the entry of data packets into the network, enabling a better use of a shared network infrastructure and avoiding congestive collapse.
- **Congestive-avoidance algorithms** (CAA) are implemented at the TCP layer as the mechanism to avoid congestive collapse in a network.
- There are two congestion control algorithms which are as follows:
 1. **Leaky Bucket Algorithm**
 2. **Token Bucket Algorithm**

Leaky Bucket Algorithm

- The leaky bucket algorithm discovers its use in the context of network traffic shaping or rate-limiting.
- A leaky bucket execution and a token bucket execution are predominantly used for traffic shaping algorithms.
- This algorithm is used to control the rate at which traffic is sent to the network and shape the burst traffic to a steady traffic stream.
- The disadvantages compared with the leaky-bucket algorithm are the inefficient use of available network resources.
- The large area of network resources such as bandwidth is not being used effectively.

Let us consider an example to understand Imagine a bucket with a small hole in the bottom. No matter at what rate water enters the bucket, the outflow is at constant rate. When the bucket is full with water additional water entering spills over the sides and is lost.



Similarly, each network interface contains a leaky bucket and the following steps are involved in leaky bucket algorithm:

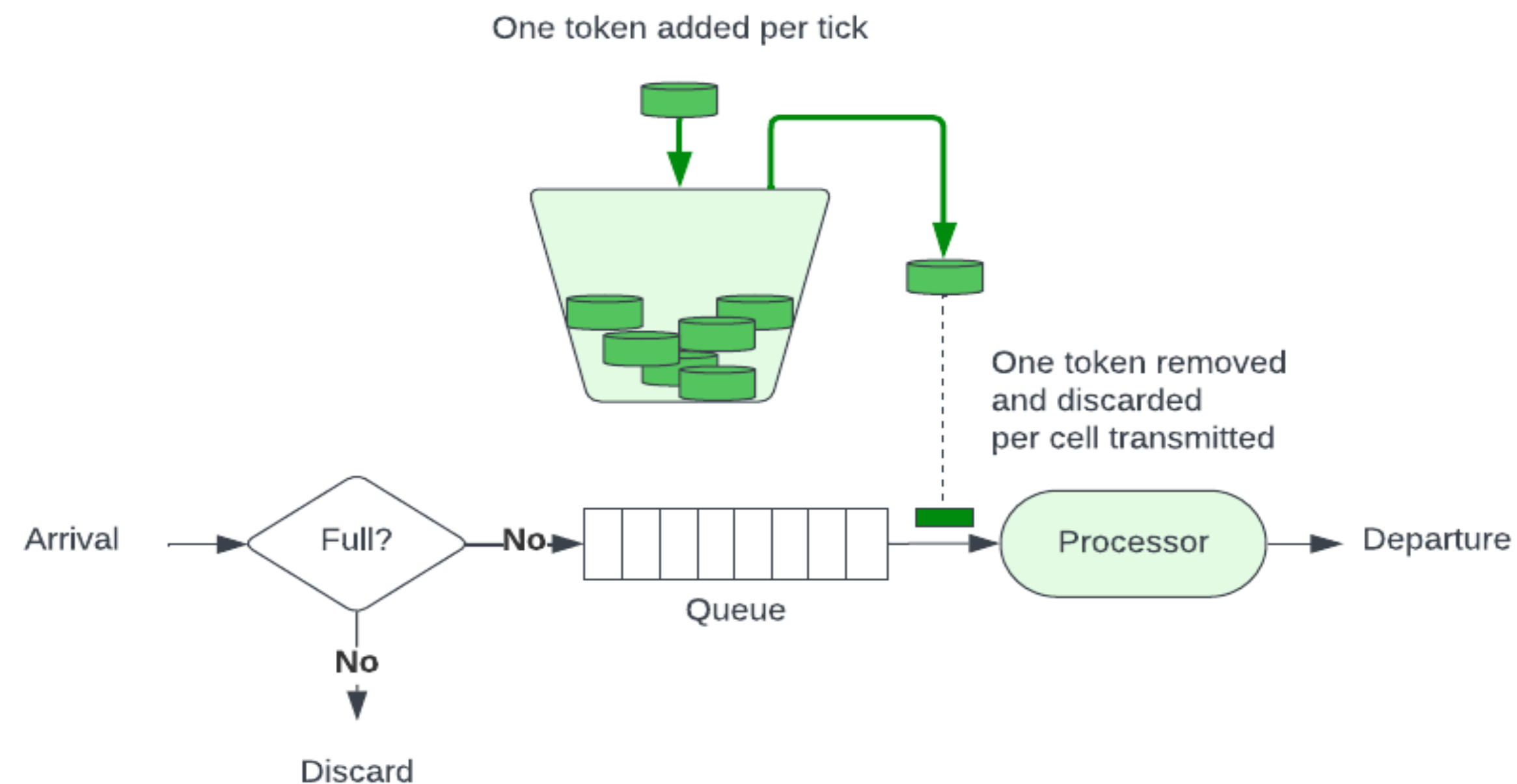
- When host wants to send packet, packet is thrown into the bucket.
- The bucket leaks at a constant rate, meaning the network interface transmits packets at a constant rate.
- Bursty traffic is converted to a uniform traffic by the leaky bucket.
- In practice the bucket is a finite queue that outputs at a finite rate.

Token Bucket Algorithm

- Leaky bucket algorithm shapes bursty traffic into fixed-rate traffic by averaging the data rate.
- It may drop the packets if the bucket is full. But this technique is very restrictive.
- It does not credit an idle host. For example, if a host is not sending for a while, its bucket becomes empty.
- If the host has bursty data, the leaky bucket allows only an average rate.
- The time when the host is idle is not taken into account. On the other hand, token bucket algorithm allows idle hosts to accumulate credit for the future in the form of tokens. And that is how it overcomes the shortcoming of leaky bucket algorithm.

Working of Token Bucket Algorithm:

- It allows bursty traffic at a regulated maximum rate. It allows idle hosts to accumulate credit for the future in the form of tokens.
- The system removes one token for every cell of data sent. For each tick of the clock the system send n tokens to the bucket. If n is 100 and host is idle for 100 ticks, bucket collects 10000 tokens. Host can now consume all these tokens with 10 cells per tick.
- Token bucket can be easily implemented with a counter. The token is initiated to zero. Each time a token is added, counter is incremented to 1. Each time a unit of data is sent, counter is decremented by 1. When the counter is zero, host cannot send data.



Process depicting how token bucket algorithm works

Steps Involved in Token Bucket Algorithm:

Step 1: Creation of Bucket: An imaginative bucket is assigned a fixed capacity, known as “rate limit”. It can hold up to a certain number of tokens.

Step 2: Refill the Bucket: The bucket is dynamic; it gets periodically filled with tokens. Tokens are added to the bucket at a fixed rate.

Step 3: Incoming Requests: Upon receiving a request, we verify the presence of tokens in the bucket.

Step 4: Consume Tokens: If there are tokens in the bucket, we pick one token from it. This means the request is allowed to proceed. The time of token consumption is also recorded.

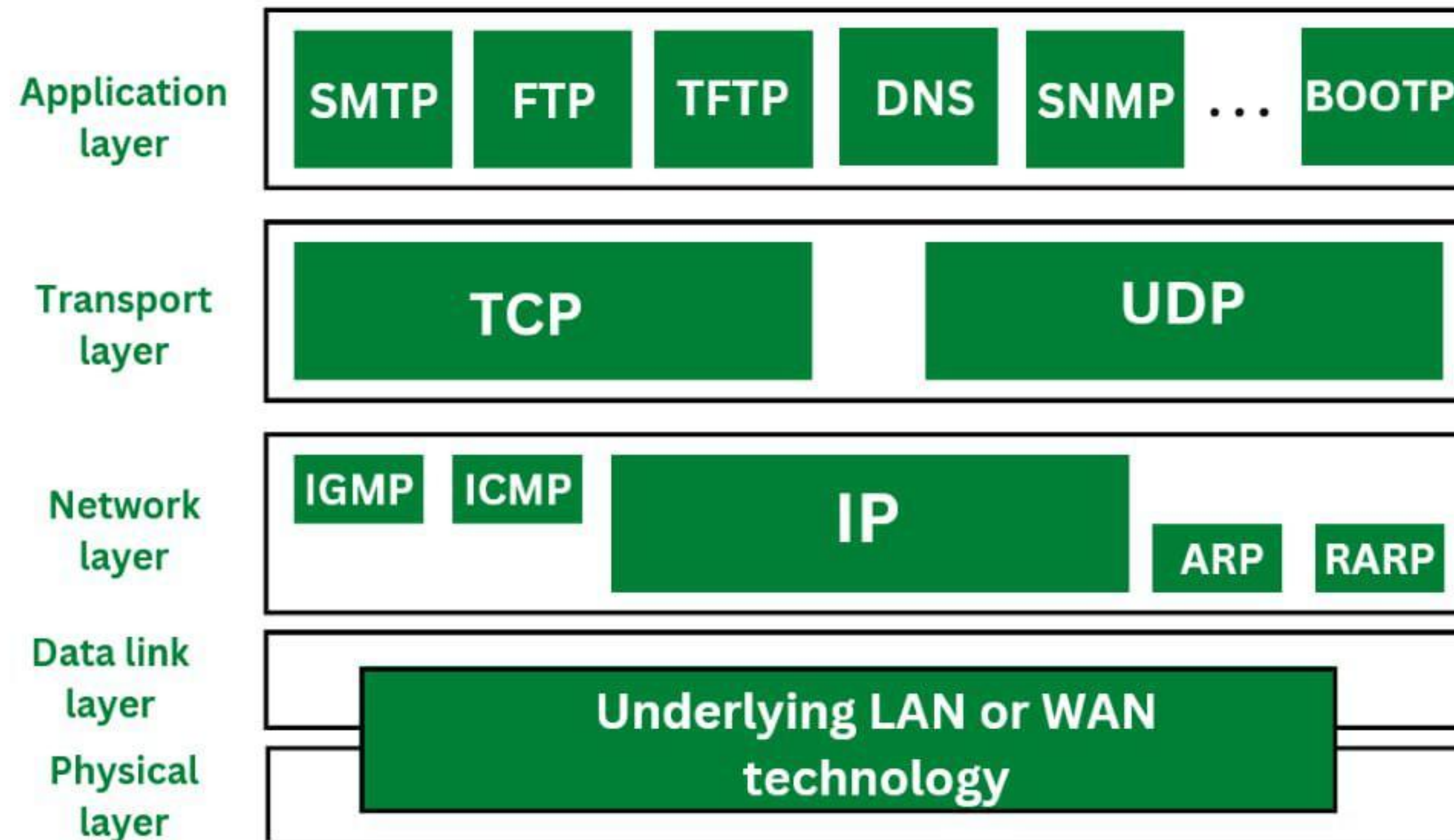
Step 5: Empty Bucket: If the bucket is depleted, meaning there are no tokens remaining, the request is denied. This precautionary measure prevents server or system overload, ensuring operation stays within predefined limits.

Advantage of Token Bucket over Leaky Bucket:

- If a bucket is full in tokens, then tokens are discarded and not the packets. While in leaky bucket algorithm, **packets are discarded**.
- Token bucket can send large bursts at a **faster rate** while leaky bucket always sends packets at constant rate.
- Token bucket ensures **predictable traffic shaping** as it allows for setting token arrival rate and maximum token count. In leaky bucket, such control may not be present.
- Premium Quality of Service(QoS) is provided by prioritizing different traffic types through distinct token arrival rates. Such **flexibility in prioritization** is not provided by leaky bucket.
- Token bucket is suitable for high-speed data transfer or streaming video content as it allows **transmission of large bursts of data**. As leaky bucket operates at a constant rate, it can lead to less efficient bandwidth utilization.
- Token Bucket provides more **granular control** as administrators can adjust token arrival rate and maximum token count based on network requirements. Leaky Bucket has limited granularity in controlling traffic compared to Token Bucket.

Transport Layer Protocols

The transport layer is the fourth layer in the OSI model and the second layer in the TCP/IP model. The transport layer provides with end to end connection between the source and the destination and reliable delivery of the services. Therefore transport layer is known as the end-to-end layer. The transport layer takes the services from its upward layer which is the application layer and provides it to the network layer. Segment is the unit of data encapsulation at the transport layer.

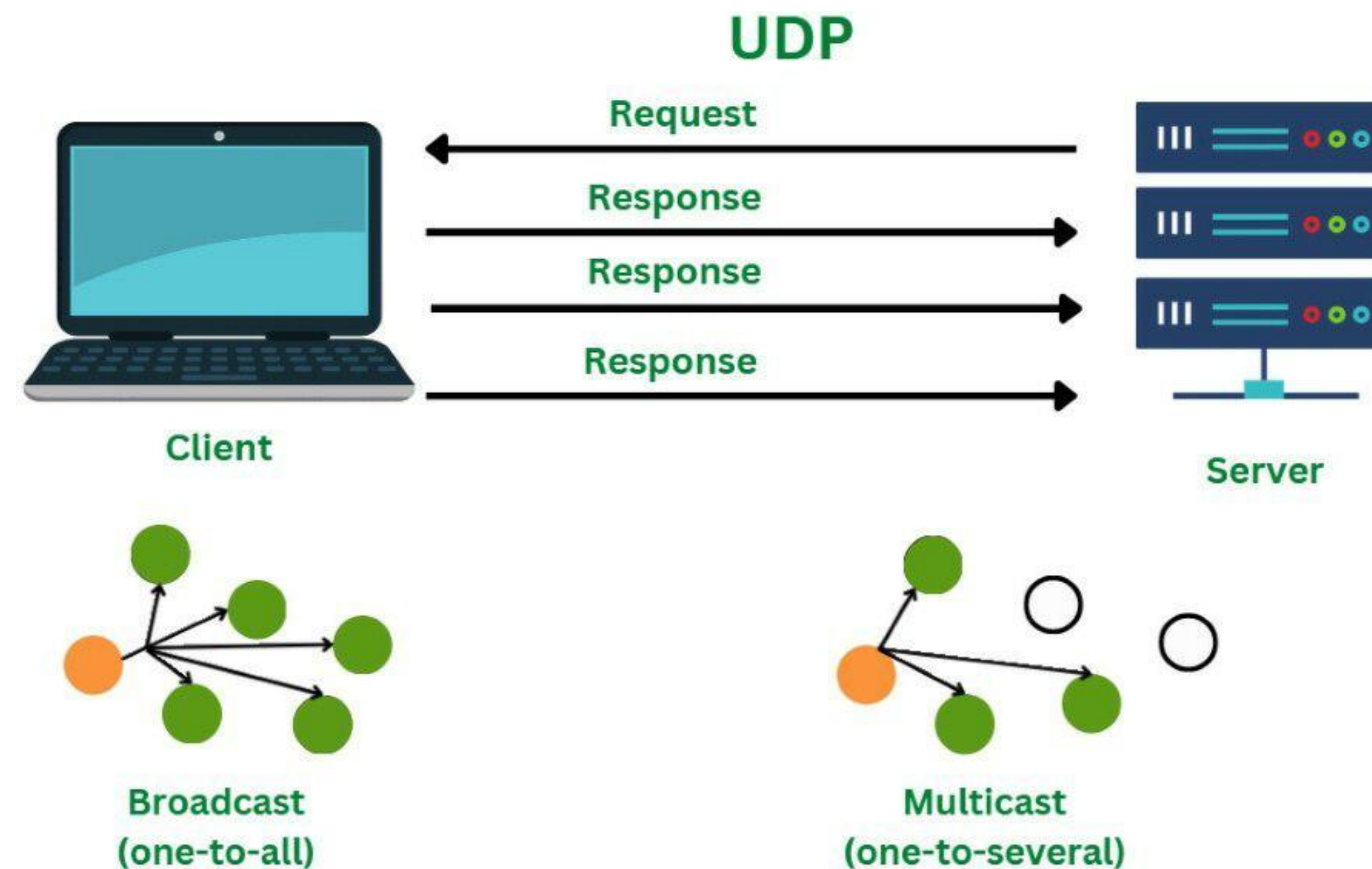


Transport Layer Protocols:

The transport layer is represented majorly by TCP and UDP protocols. Today almost all operating systems support multiprocessing multi-user environments. This transport layer protocol provides connections to the individual ports. These ports are known as protocol ports. Transport layer protocols work above the IP protocols and deliver the data packets from IP serves to destination port and from the originating port to destination IP services. Below are the protocols used at the transport layer.

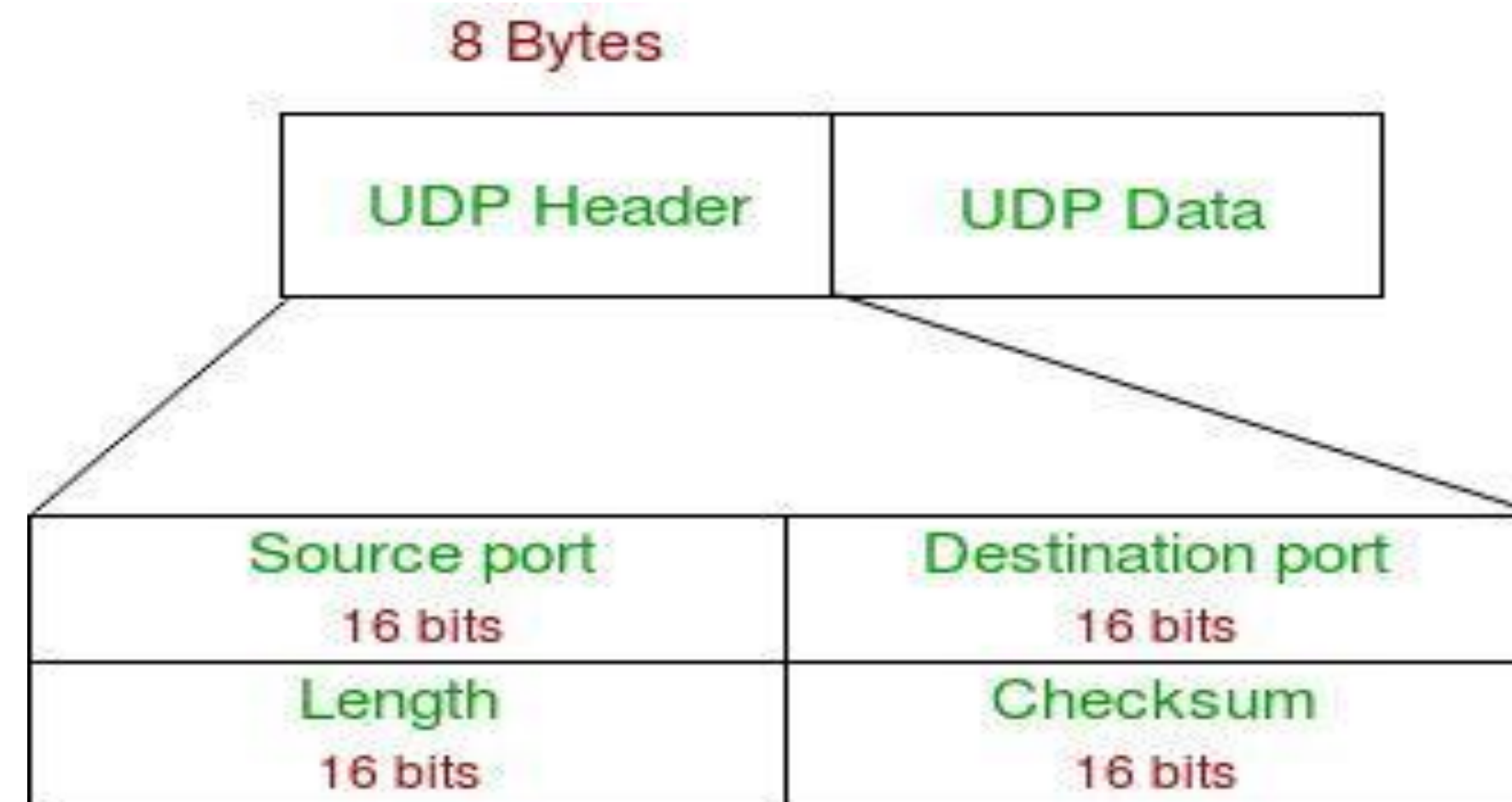
1. UDP

UDP stands for User Datagram Protocol. User Datagram Protocol provides a nonsequential transmission of data. It is a connectionless transport protocol. UDP protocol is used in applications where the speed and size of data transmitted is considered as more important than the security and reliability. User Datagram is defined as a packet produced by User Datagram Protocol. UDP protocol adds checksum error control, transport level addresses, and information of length to the data received from the layer above it. Services provided by User Datagram Protocol(UDP) are connectionless service, faster delivery of messages, checksum, and process-to-process communication.



UDP Segment

While the TCP header can range from 20 to 60 bytes, the UDP header is a fixed, basic 8 bytes. All required header information is contained in the first 8 bytes, with data making up the remaining portion. Because UDP port number fields are 16 bits long, the range of possible port numbers is defined as 0 to 65535, with port 0 being reserved.



UDP

- Source Port:** Source Port is a 2 Byte long field used to identify the port number of the source.
- Destination Port:** This 2-byte element is used to specify the packet's destination port.
- Length:** The whole length of a UDP packet, including the data and header. The field has sixteen bits.
- Checksum:** The checksum field is two bytes long. The data is padded with zero octets at the end (if needed) to create a multiple of two octets. It is the 16-bit one's complement of the one's complement sum of the UDP header, the pseudo-header containing information from the IP header, and the data.

Advantages of UDP

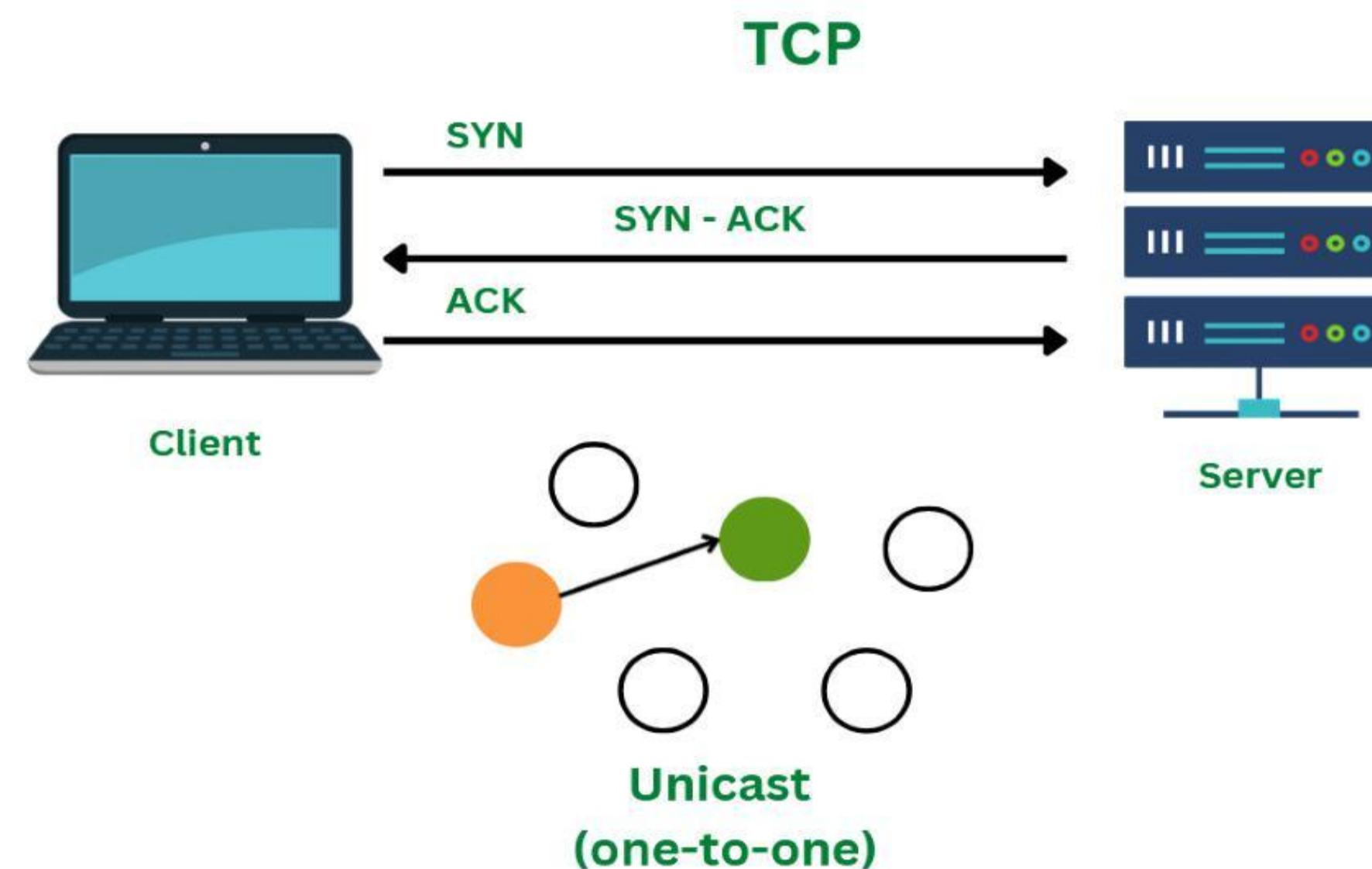
- UDP also provides multicast and broadcast transmission of data.
- UDP protocol is preferred more for small transactions such as DNS lookup.
- It is a connectionless protocol, therefore there is no compulsion to have a connection-oriented network.
- UDP provides fast delivery of messages.

Disadvantages of UDP

- In UDP protocol there is no guarantee that the packet is delivered.
- UDP protocol suffers from worse packet loss.
- UDP protocol has no congestion control mechanism.
- UDP protocol does not provide the sequential transmission of data.

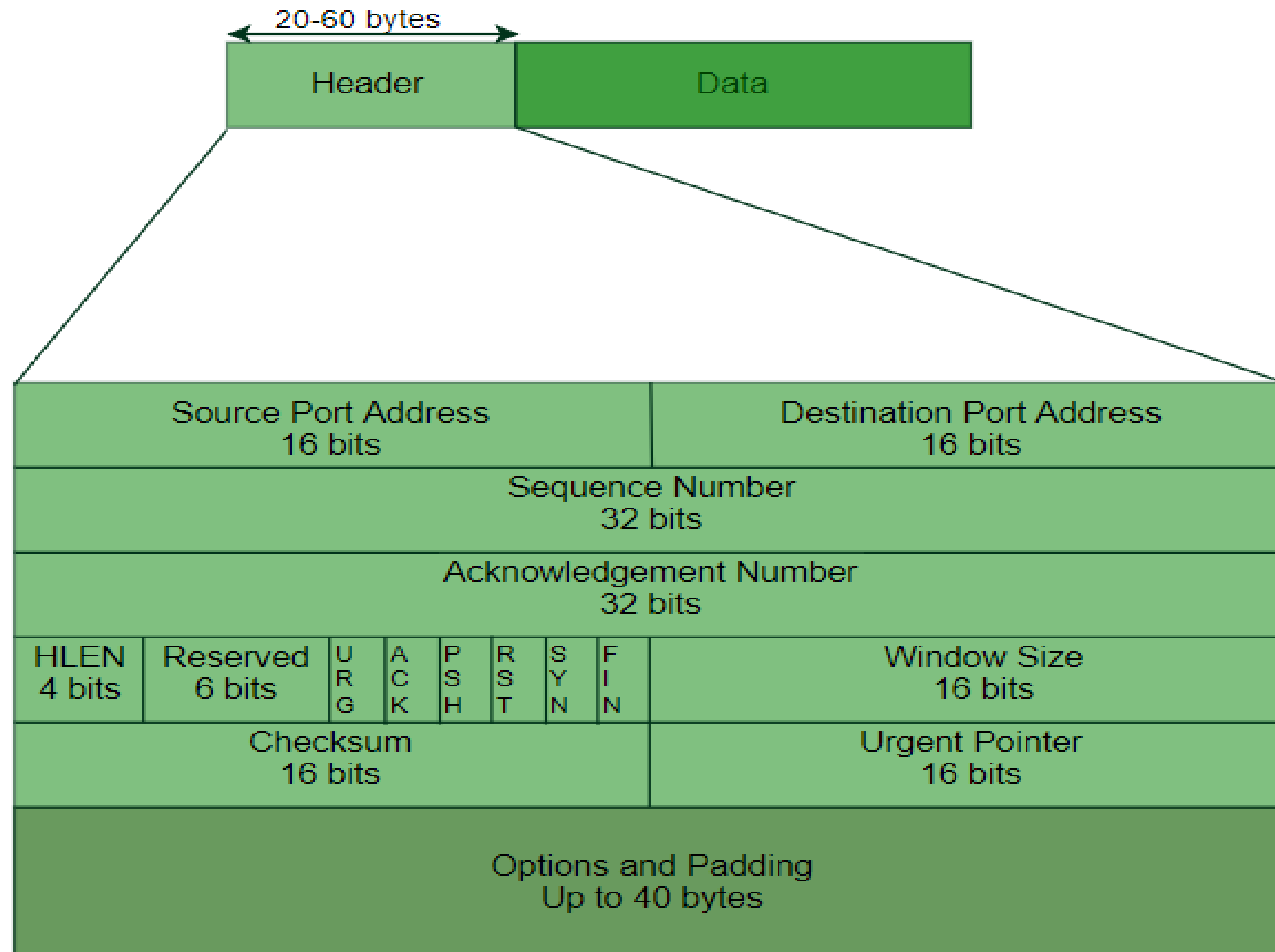
2. TCP

- TCP stands for Transmission Control Protocol. TCP protocol provides transport layer services to applications.
- TCP protocol is a connection-oriented protocol. A secured connection is being established between the sender and the receiver. For a generation of a secured connection, a virtual circuit is generated between the sender and the receiver.
- The data transmitted by TCP protocol is in the form of continuous byte streams.
- A unique sequence number is assigned to each byte. With the help of this unique number, a positive acknowledgment is received from receipt.
- If the acknowledgment is not received within a specific period the data is retransmitted to the specified destination.



TCP Segment:

A TCP segment's header may have 20–60 bytes. The options take about 40 bytes. A header consists of 20 bytes by default, although it can contain up to 60 bytes.



- Source Port Address:** The port address of the programme sending the data segment is stored in the 16-bit field known as the source port address.
- Destination Port Address:** The port address of the application running on the host receiving the data segment is stored in the destination port address, a 16-bit field.
- Sequence Number:** The sequence number, or the byte number of the first byte sent in that specific segment, is stored in a 32-bit field. At the receiving end, it is used to put the message back together once it has been received out of sequence.
- Acknowledgement Number :** The acknowledgement number, or the byte number that the recipient anticipates receiving next, is stored in a 32-bit field called the acknowledgement number. It serves as a confirmation that the earlier bytes were successfully received.
- Header Length (HLEN):** This 4-bit field stores the number of 4-byte words in the TCP header, indicating how long the header is. For example, if the header is 20 bytes (the minimum length of the TCP header), this field will store 5 because $5 \times 4 = 20$, and if the header is 60 bytes (the maximum length), it will store 15 because $15 \times 4 = 60$. As a result, this field's value is always between 5 and 15.

- **Control flags:** These are six 1-bit control bits that regulate flow control, method of transfer, connection abortion, termination, and establishment. They serve the following purposes:
 - **Urgent:** This pointer is legitimate
 - **ACK:** The acknowledgement number (used in cumulative acknowledgement cases) is valid.
 - **PSH:** Push request
 - **RST:** Restart the link.
 - **SYN:** Sequence number synchronisation
 - **FIN:** Cut off the communication
 - **Window size:** This parameter provides the sender TCP's window size in bytes.
- **Checksum:** The checksum for error control is stored in this field. Unlike UDP, it is required for TCP.
- **Urgent pointer:** This field is used to point to data that must urgently reach the receiving process as soon as possible. It is only valid if the URG control flag is set. To obtain the byte number of the final urgent byte, the value of this field is appended to the sequence number.

Advantages of TCP

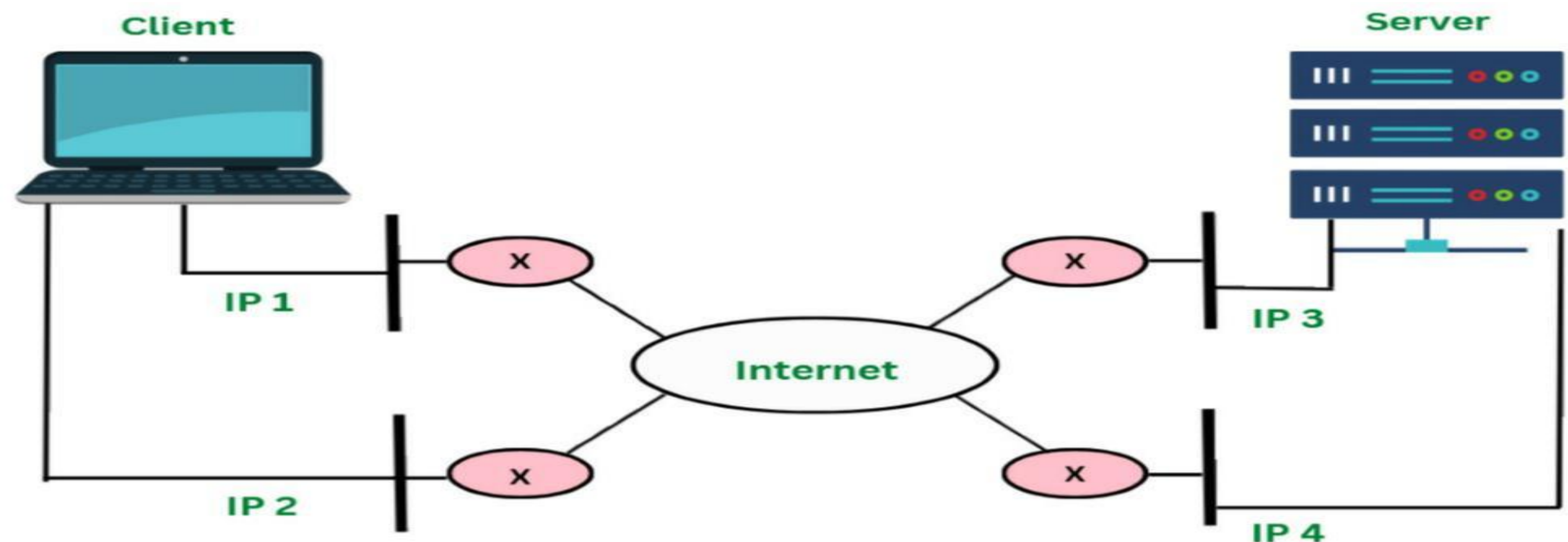
- TCP supports multiple routing protocols.
- TCP protocol operates independently of that of the operating system.
- TCP protocol provides the features of error control and flow control.
- TCP provides a connection-oriented protocol and provides the delivery of data.

Disadvantages of TCP

- TCP protocol cannot be used for broadcast or multicast transmission.
- TCP protocol has no block boundaries.
- No clear separation is being offered by TCP protocol between its interface, services, and protocols.
- In TCP/IP replacement of protocol is difficult.

3. SCTP

- SCTP stands for Stream Control Transmission Protocol. SCTP is a connection-oriented protocol. Stream Control Transmission Protocol transmits the data from sender to receiver in full duplex mode.
- SCTP is a unicast protocol that provides with point to point-to-point connection and uses different hosts for reaching the destination.
- SCTP protocol provides a simpler way to build a connection over a wireless network. SCTP protocol provides a reliable transmission of data. SCTP provides a reliable and easier telephone conversation over the internet.
- SCTP protocol supports the feature of multihoming ie. it can establish more than one connection path between the two points of communication and does not depend on the IP layer.



Advantages of SCTP

- SCTP provides a full duplex connection. It can send and receive the data simultaneously.
- SCTP protocol possesses the properties of both TCP and UDP protocol.
- SCTP protocol does not depend on the IP layer.
- SCTP is a secure protocol.

Disadvantages of SCTP

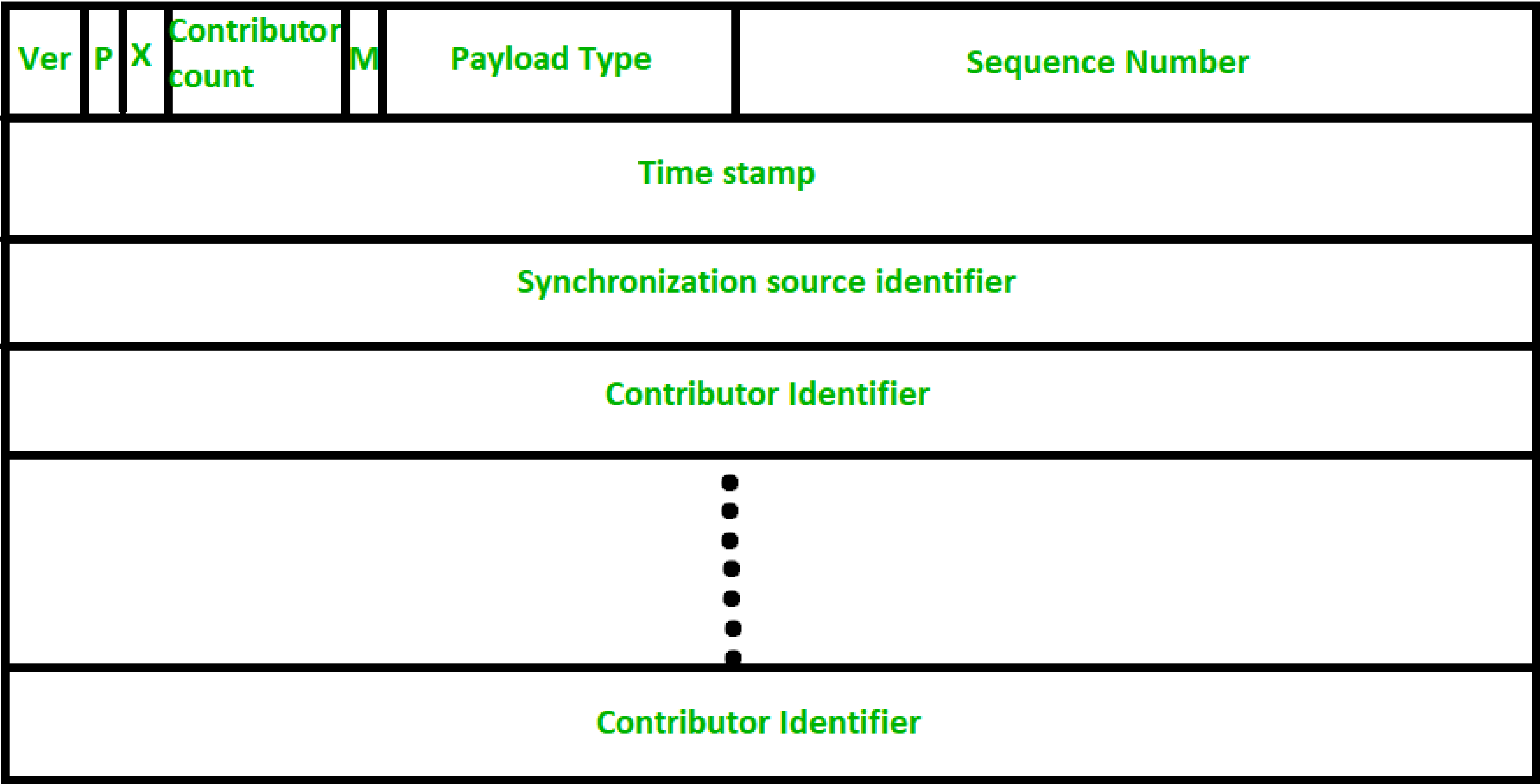
- To handle multiple streams simultaneously the applications need to be modified accordingly.
- The transport stack on the node needs to be changed for the SCTP protocol.
- Modification is required in applications if SCTP is used instead of TCP or UDP protocol.

Real Time Transport Protocol (RTP)

A protocol is designed to handle real-time traffic (like audio and video) of the Internet, is known as **Real Time Transport Protocol (RTP)**. RTP must be used with UDP. It does not have any delivery mechanism like multicasting or port numbers. RTP supports different formats of files like MPEG and MJPEG. It is very sensitive to packet delays and less sensitive to packet loss.

- 1.RTP mainly helps in media mixing, sequencing and time-stamping.
- 2.Voice over Internet Protocol (VoIP)
- 3.Video Conferencing over Internet.
- 4.Internet Audio and video streaming.

RTP Header Format : The diagram of header format of RTP packet is shown below:



The header format of RTP is very simple and it covers all real-time applications. The explanation of each field of header format is given below:

- Version** : This 2-bit field defines version number. The current version is 2.
- P** – The length of this field is 1-bit. If value is 1, then it denotes presence of padding at end of packet and if value is 0, then there is no padding.
- X** – The length of this field is also 1-bit. If value of this field is set to 1, then it indicates an extra extension header between data and basic header and if value is 0 then, there is no extra extension.
- Contributor count** – This 4-bit field indicates number of contributors. Here maximum possible number of contributor is 15 as a 4-bit field can allow number from 0 to 15.
- M** – The length of this field is 1-bit and it is used as end marker by application to indicate end of its data.
- Payload types** – This field is of length 7-bit to indicate type of payload. We list applications of some common types of payload.
- Sequence Number** – The length of this field is 16 bits. It is used to give serial numbers to RTP packets. It helps in sequencing. The sequence number for first packet is given a random number and then every next packet's sequence number is incremented by 1. This field mainly helps in checking lost packets and order mismatch.

- **Time Stamp** – The length of this field is 32-bit. It is used to find relationship between times of different RTP packets. The timestamp for first packet is given randomly and then time stamp for next packets given by sum of previous timestamp and time taken to produce first byte of current packet. The value of 1 clock tick is varying from application to application.
- **Synchronization Source Identifier** – This is a 32-bit field used to identify and define the source. The value for this source identifier is a random number that is chosen by source itself. This mainly helps in solving conflict arises when two sources started with the same sequencing number.
- **Contributor Identifier** – This is also a 32-bit field used for source identification where there is more than one source present in session. The mixer source use Synchronization source identifier and other remaining sources (maximum 15) use Contributor identifier.

Quality of Service(QoS)

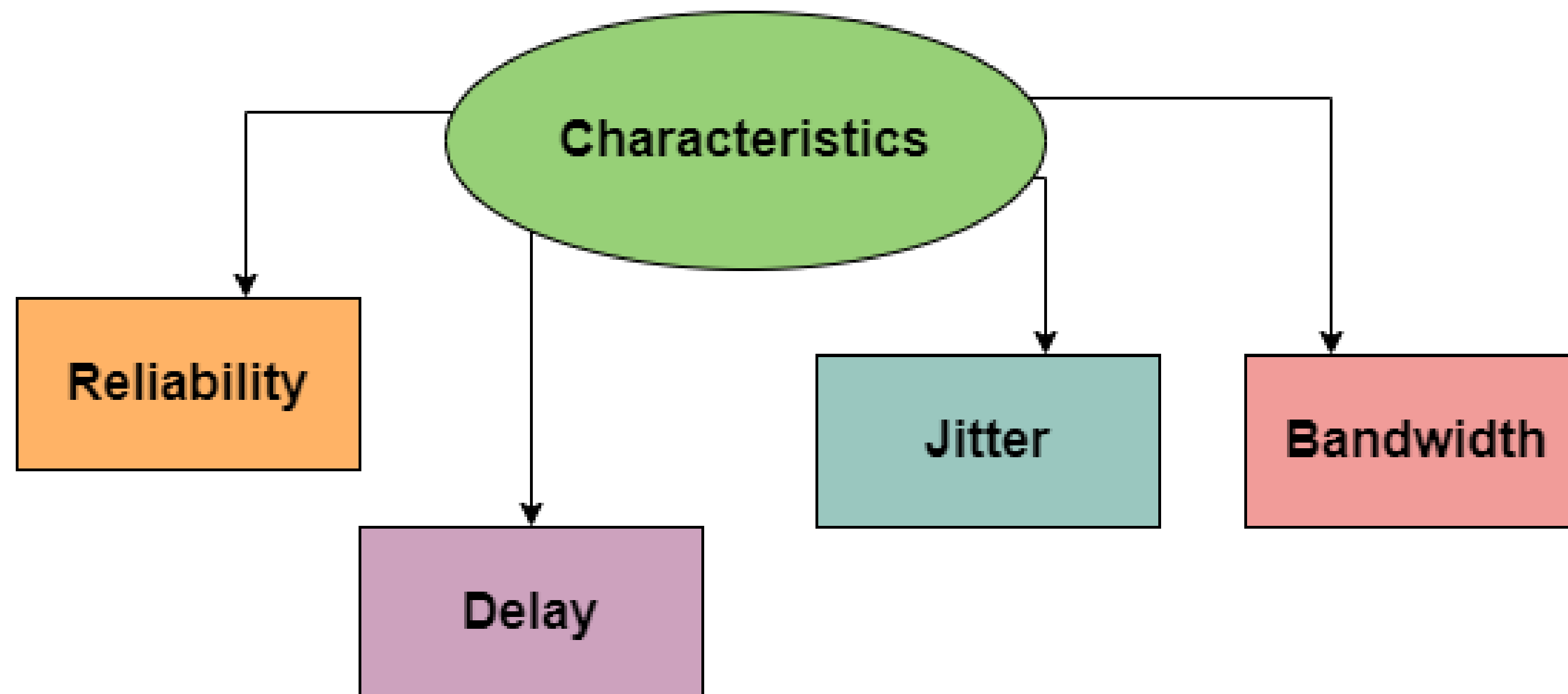
Quality of Service(QoS) is basically the ability to provide different priority to different applications, users, or data flows, or in order to guarantee a certain level of performance to the flow of data.

QoS is basically the overall performance of the computer network. Mainly the performance of the network is seen by the user of the Network.

Flow Characteristics

Given below are four types of characteristics that are mainly attributed to the flow and these are as follows:

- Reliability
- Delay
- Jitter
- Bandwidth



Reliability

- It is one of the main characteristics that the flow needs. If there is a lack of reliability then it simply means losing any packet or losing an acknowledgement due to which retransmission is needed.
- Reliability becomes more important for electronic mail, file transfer, and for internet access.

Delay

- Another characteristic of the flow is the delay in transmission between the source and destination. During audio conferencing, telephony, video conferencing, and remote conferencing there should be a minimum delay.

Jitter

- It is basically the variation in the delay for packets that belongs to the same flow. Thus Jitter is basically the variation in the packet delay. Higher the value of jitter means there is a large delay and the low jitter means the variation is small.

Bandwidth

- The different applications need different bandwidth.

How to achieve Quality of Service:

Let's get into some details and say, your organization wants to achieve Quality of Service, which can be done by using some tools and techniques, like **jitter buffer** and **traffic shaping**.

Jitter buffer:

- This is a **temporary storage buffer** which is used to store the incoming data packets, it is used in **packet-based networks** to ensure that the **continuity of the data streams** doesn't get disturbed, it does that by **smoothing out the packet arrival** times during periods of network congestion.

Traffic shaping:

- This technique which is also known as **packet shaping** is a **congestion control or management technique** that helps to regulate network data transfer by delaying the flow of least important or least necessary data packets.
- QoS is included in the service-level agreement when an organization signs it with its network service provider which guarantees the selected performance level.

There are 2 types of Quality of Service Solutions:

1. Stateless solution: Here, the server is not required to keep or store the server information or session details to itself. The **routers** maintain no fine-grained state about traffic, one positive factor of this is, that it's **scalable and robust**. But also, it has weak services as there is **no guarantee about the kind of performance delay** in a particular application which we encounter. In the stateless solution, the server and client are **loosely coupled** and can act.

2. Stateful solution: Here, the server is required to maintain the **current state and session information**, the **routers maintain per-flow state** as the flow is very important in providing the Quality-of-Service which is providing powerful services such as guaranteed services and high resource utilization, provides protection, and is much **less scalable and robust**. Here, the server and client are **tightly bounded**.

Quality of Service Parameters:

QoS can be measured quantitatively by using several parameters

- Packet loss:** it happens when the network links become congested and the routers and switches start dropping the packets. When these packets are dropped during real-time communication, such as audio or video, these sessions can experience jitter and gaps in speech.
- Jitter:** occurs as the result of network congestion, timing drift, and route changes. And also, too much jitter can degrade the quality of audio communication.
- Latency:** is the time delay, which is taken by a packet to travel from its source to its destination. For a great system, latency should be as low as possible, ideally, it should be close to zero.
- Bandwidth:** is the capacity of a network channel to transmit maximum possible data through the channel in a certain amount of time. QoS optimizes a network by managing its bandwidth and setting the priorities for those applications which require more resources as compared to other applications.
- Mean opinion score:** it is a metric for rating the audio quality which uses a five-point scale, with a five indicating the highest or best quality.

Implementing Quality of Service:

We can implement Quality of service through three of the following existing models:

1.Best Effort: if we are applying this model then, it means that we are prioritizing all the data packets equally. But since we all setting the priority order like this, then there is no guarantee that all the data packets will be delivered, but it will put up the best effort to deliver all of them. Point to remember is, that the best-effort model is applied when networks haven't configured with the QoS policies or incase their network infrastructure does not support QoS.

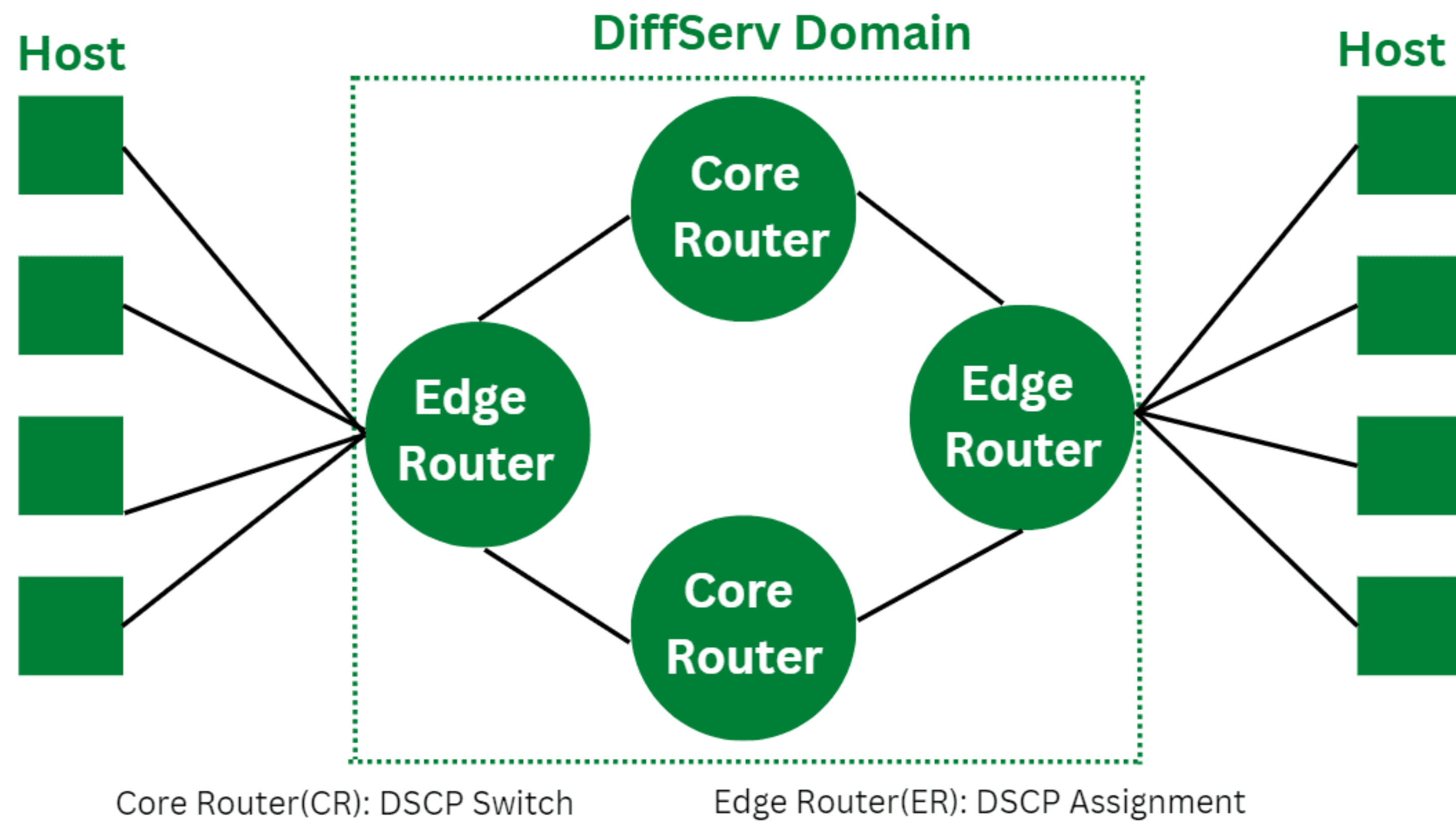
2.Integrated Services: or IntServ, this QoS model reserves the bandwidth along a specific path on the network. The applications ask the network's resource reservation for themselves and parallelly the network devices monitor the flow of packets to make sure network resources can accept packets. Point to remember: while implementing Integrated Services Model, the IntServ-capable routers and resource reservation protocol are necessary. This model has limited scalability and high consumption of the network resources.

3.Differentiated Services: in this QoS model, the network elements such as routers and switches are configured to serve multiple categories of traffic with different priority orders. A company can categorize the network traffic based on its requirements. Eg. Assigning higher priority to audio traffic etc.

Differentiated services:

Differentiated Services (DiffServ) is defined as a class of service (COS) model that is used to describe and control the IP network traffic by class. The main aim of differentiated services is to give priority to the specific traffic that needs an uninterrupted flow of data. In Differentiated services, the traffic is divided into multiple classes and each class is treated and prioritized differently. This technique of classification used by differentiated services is useful when there are limited or less number of resources. Differentiated services work at the Network layer of the Open Systems Interconnection (OSI) Model. An example of Differentiated services is voice traffic. DiffServ Domain is used to identify the scope of the DiffServ protocol. The domain consists of two nodes. They are:

- . Edge Routers
- . Core Routers



1. Edge Routers: Edge routers are located at the borders of the domain. The functions performed by edge routers are:

- . To horizontally map the DiffServ and QoS protocol that is inside the incoming domain.
- . To assign the same DSCP values to the traffic flows that have same QoS requirements.
- . To assign the DSCP value to all of the packets according to the QoS information stated by the incoming network.

2. Core Routers: Core routers are present inside the domain. The functions performed by core routers are:

- . To forward the packets that have different policies according to their DSCP values.
- . To make sure that DSCP values are treated similarly inside the domain.

The traffic classes are grouped into three types by DiffServ protocol. They are as follows:

1. Expedited Forwarding (EF): Expedited Forwarding (EF) identifies the traffic with specific QoS requirements like telephony.

2. Assured Forwarding (AF): Assured Forwarding (AF) consists of traffic classes such as AF1, AF2, AF3 and AF4.

3. Best Effort (BE): Best Effort (BE) does not assure about QoS.

Advantages of Differentiated Services (DiffServ)

- › There is no need for Differentiated services to maintain the soft state and refresh each reserved node periodically to recover from errors.
- › Differentiated services are more scalable as the allocation of resources takes place in the granularity of class.
- › Differentiated services can simplify network operations.
- › Differentiated services are used where there is a need to provide prioritized service mechanisms.

Disadvantages of Differentiated Services (DiffServ)

- › Differentiated services do not provide guaranteed performance.
- › Differentiated services work more effectively only when the high-priority traffic is less.